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Science understanding

BIOLOGY FOR YEARS 7–8

Beth Montgomery

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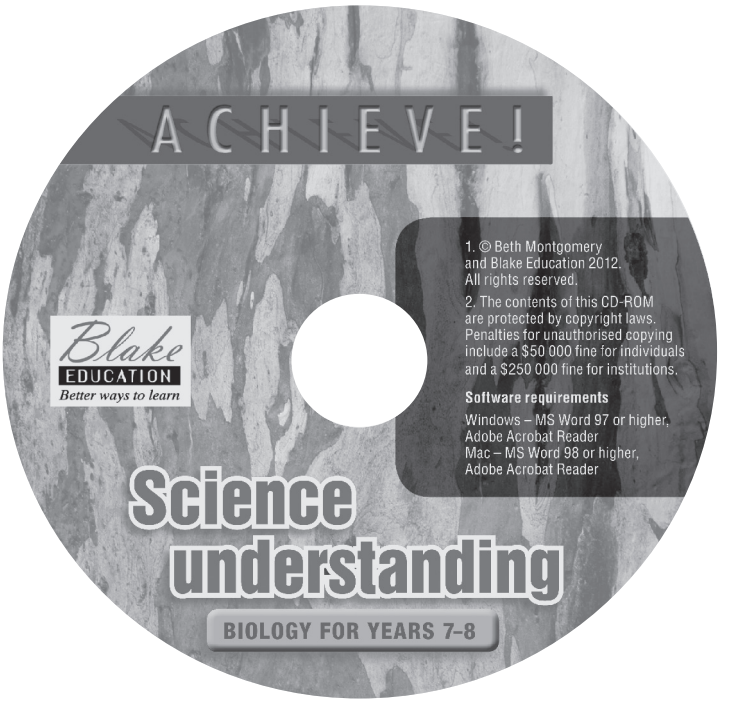

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- ✓ Emphasis on understanding essential science content and concepts.
- ✓ Comprehensive Teacher's Notes that provide clear objectives, background information, starter and extension activities.
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Software requirements

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- ✓ Word files on the CD-ROM that can be edited to suit the specific needs of students.
- ✓ All illustrations are included on the CD-ROM and can be used on new teacher-created work sheets.

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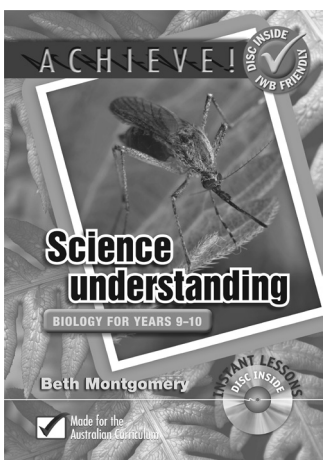

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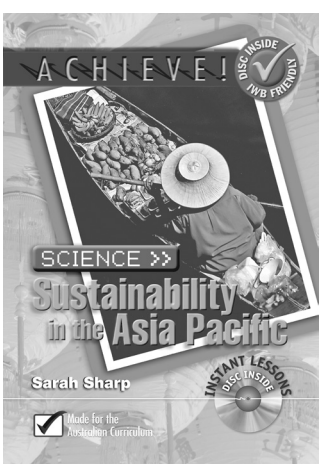

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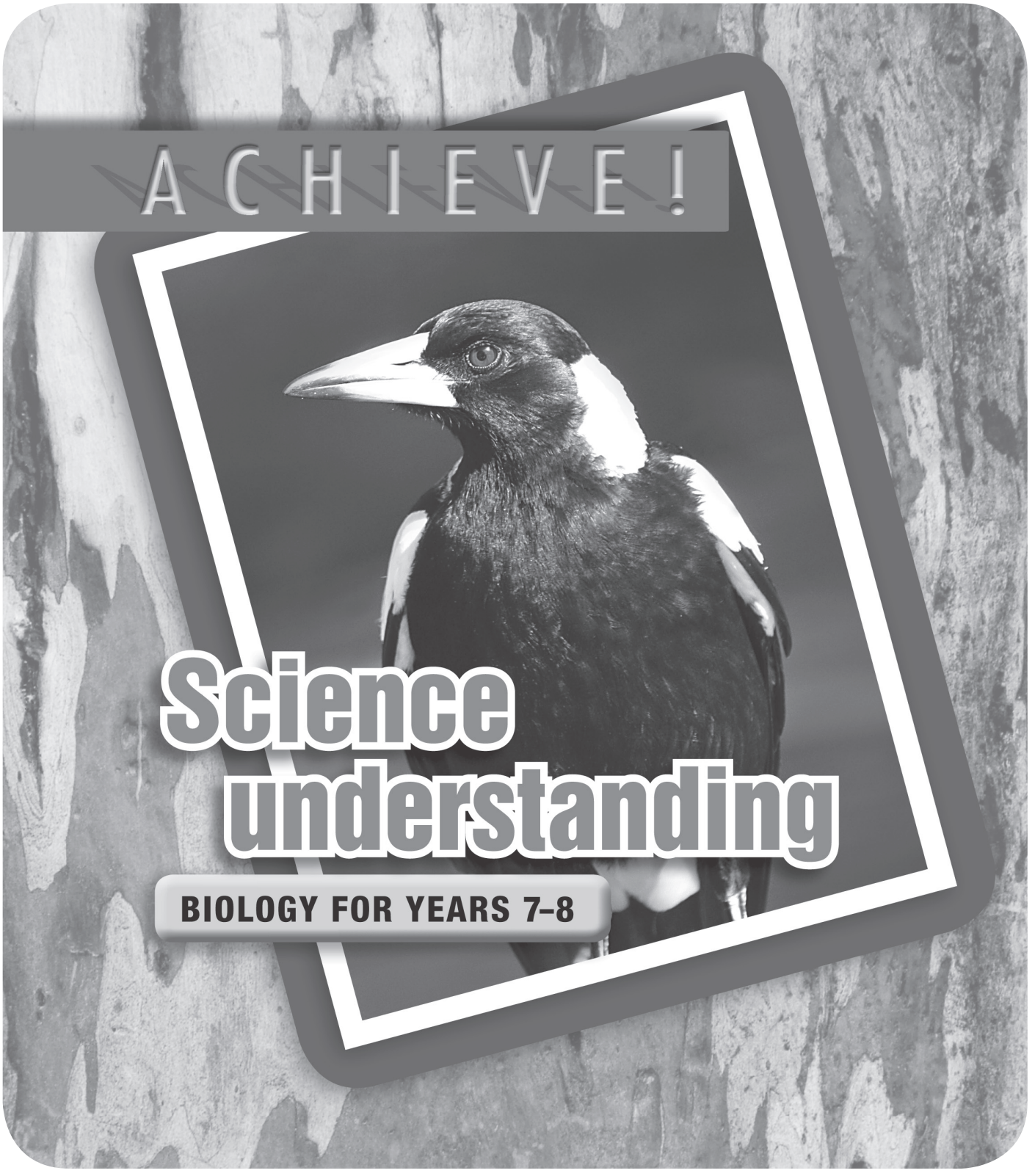

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Introduction

Achieve!

The *Achieve!* series of Science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

This title will assist teachers to develop lessons for the biology sub-strand in *Science understanding* for years 7 and 8. Units of work cover a variety of topics including classifying organisms according to characteristics, vertebrates and invertebrates, food chains and food webs and the role of humans and introduced species in these, cell structures and analysis, and systems such as the digestive system, reproductive system, eyes and ears. Work sheets help students build their scientific language knowledge and understanding of key concepts.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the individual needs of students.

This book is divided into ten units. The units can be taught as a lesson with students in groups of varying sizes. Alternatively, a single Activity sheet can be used as support material, for individual homework, assessment or revision.

The teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the work sheets.

Prior knowledge – This refers to the skills and knowledge required for students to complete the tasks. Some activities are more challenging than others, however, as a rule, students should have a reading comprehension age of six to nine years to be able to attempt the activities.

Background – This section provides additional guidance to help teachers to present a successful lesson. It may include definitions or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Activity and Resource work sheets – The Resource sheets do not always contain activities and are used either to stimulate discussion or as a part of the unit's activities. The Activity work sheets may be used sequentially or as stand-alone tasks, depending on the needs of the student.

Extension ideas – Additional tasks are suggested to recap on the main skills or knowledge taught in the lesson, as well as ways to extend, assess or include technology in the lesson.

The Australian Curriculum correlations

	Year 7 / Year 8	Units 1	2	3	4	5	6	7	8	9	10
Science understanding Biological sciences											
There are differences within and between groups of organisms; classification helps organise this diversity	ACSSU111	✓	✓								
Interactions between organisms can be described in terms of food chains and food webs; human activity can affect these interactions	ACSSU112			✓	✓						
Cells are the basic units of living things and have specialised structures and functions	ACSSU149					✓	✓				
Multicellular organisms contain systems of organs that carry out specialised functions that enable them to survive and reproduce	ACSSU150							✓	✓	✓	✓

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Teacher's notes – Classification basics

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Objectives

- To understand that organisms are sorted according to characteristics
- To be able to sort organisms according to characteristics
- To know that the binomial system is used to name organisms
- To be aware that extinct organisms are also classified and named

Prior knowledge

Students should have a basic understanding of how some creatures are similar to one another in structure and shape. For example, saltwater crocodiles/freshwater crocodiles/alligators, or cats/lions/tigers. Students should be able to list differences in characteristics, such as how a crocodile is different to a cat.

Background

Scientists sort all living and extinct organisms into groups, classifying them according to their characteristics. This cataloguing process ensures no confusion exists about an organism's identity. Scientists have developed various classification methods over the last 300 years but have kept to a hierarchical system based on the work of Swedish botanist Carl Linnaeus.

Starter activity

As a class, brainstorm a list of animals or use photos or wildlife calendar pictures. Ask students to identify which animals can be grouped together as having similar characteristics and why.

Activity sheets

Scientists need to be able to identify a species properly. The Activity sheet 'Which cabbage?' illustrates the different characteristics they take into account when doing this. Ask students what vegetables they know, if any. Show specimens, and list ambiguous vegetables such as peppers/capsicum, marrow/zucchini. Add any foreign language names if ESL students are present. Ask students to decide which name is correct. Introduce the concept of scientific names.

'The platypus – bird, mammal or reptile?' Activity sheet gives some history of classification and introduces students to monotremes. Have students label the diagram and match the statements about platypus characteristics. Discuss the differences between mammals, reptiles and birds so students can answer the higher order questions.

The Activity sheet 'The seven levels of classification' explains that organisms are sorted into a hierarchy of groups. Students can use the word bank to fill out the table. Encourage students not to be intimidated by the Greek and Latin names and explain that this is how organisms are named, regardless of what country you are in or what language you speak.

The 'Scientific naming' Activity sheet explains that the binomial (two name) system of naming organisms uses the terms 'genus' and 'species'. Students can use the notes to write a paragraph about scientific naming conventions. Then they can use the information to correctly name the animals shown.

In the Activity sheet 'Dinosaur classification', students learn that scientists aim to classify all species, even if a species is extinct. Dinosaurs and other prehistoric creatures have all been classified. Students use the word bank to complete the passage and then sort dinosaurs into two categories: lizard-hipped and bird-hipped.

Extension ideas

Show the class pictures of different organisms and ask them to name each organism and suggest which kingdom/phylum/order/class/family each organism belongs to. This is easiest with animal pictures, but mix it up with a few plants and some fungi if possible.

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Which cabbage?

☞ **1** Use the word bank to fill in the passage below.

Word bank

same
Australia
plant
scientific
cabbage
different
Animals

In the Pacific people eat leafy greens called c_____. In Tonga they call their cabbage *pele*. In Papua New Guinea it's called *aibika*, and in the Solomon Islands it's *slippery cabbage*. It's all exactly the same p_____ – but is it the s_____ as *Chinese cabbage*? How are these plants d_____ to the round type we eat in A_____? Farmers and scientists need to be sure which plant is which. That's why scientists give plants s_____ names – so there is no confusion. A_____ have **scientific names** too.

Hibiscus cabbage	**European cabbage**	**Chinese cabbage**
Grows to approx. 1 m	Grows to approx. 50 cm	Grows to approx. 50 cm
Thin, round stems	Short, curved stems	Long, flattened stems
Pointy leaves with slimy sap	Big round curved leaves	Long broad wrinkly leaves
Bushy shrub	Ball-shaped growth	Upright growth

Brassica oleracea

Abelmoschus manihot

Brassica rapa

2 Look at the pictures and table above. Tick the characteristics that you can see in the pictures, then match each of the pictures to one of the types of cabbage listed in the table.

3 Draw lines joining each common name to its scientific name.

Hibiscus cabbage	*Brassica rapa*
European cabbage	*Abelmoschus manihot*
Chinese cabbage	*Brassica oleracea*

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The platypus – bird, mammal or reptile?

☞ **1** Use the word bank to label the picture below.

Word bank

fur
claws
flat tail
duck-like bill

The platypus is a unique kind of mammal called a **monotreme**. Echidnas are the only other kind of monotreme. When scientists in Europe first saw the hide of a dead platypus they thought it was a joke. They had never seen anything like it before, and assumed it was a made-up animal sewn together from parts of other animals.

☞ **2** Match the following statements to describe the platypus.

The platypus has a bill	a poisonous spur on its back leg.
The male has	tail like a beaver.
The female platypus	claws like an otter.
It has a flat	temperature of about 32°C.
The young lick milk	like a duck.
They have	lays eggs.
The platypus has a low core	from their mother's belly fur.

3 How is a platypus like a reptile? _____

_____

4 How is a platypus like a bird? _____

_____

5 How is a platypus like a mammal? _____

_____

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The seven levels of classification

Traditionally, when classifying **organisms**, scientists sort them into groups on seven levels. This is known as a **hierarchical system**. Each level classifies the organism in more detail. The first, and highest, of these levels is **Kingdom**. Organisms can be in either the **animal** or **plant** kingdom, or in other kingdoms such as **fungi** and **bacteria**.

☞ **1** What kingdom would you place the following organisms in?

platypus _____ carrot _____ frog _____ rose _____

Next, organisms are sorted by their structure. This refers to whether an animal has a backbone (chordata), or a soft body or a hard outer shell. This level is called the **Phylum**.

Organisms are then sorted by **Class**, then **Order**, **Family**, **Genus** and **Species**. Each level goes into more detail about the structure, shape and other characteristics of the organism. The group names are in Latin or Greek, but they are often similar to common English names. For example, Class *mammalia* means mammals.

☞ **2** Use the word bank to fill out the table. Note, some words are used more than once.

Word bank

Insecta	Animal	Mammalia
Marsupialia	Chordata	Octopoda

	Feather-tailed glider	**Bogong moth**	**Blue-ringed octopus**
Kingdom:	A_____	A_____	A_____
Phylum:	C_____	Arthropoda	Mollusca
Class:	M_____	I_____	Cephalopoda
Order:	M_____	Lepidoptera	O_____
Family:	Acrobatidae	Noctuidae	Octopodidae
Genus:	*Acrobates*	*Agrotis*	*Hapalochlaena*
Species:	*pygmaeus*	*infusa*	*maculosa*

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Scientific naming

- **Scientific names** consist of genus and species.
- The species name is all lower case.
- Scientific names are written in Latin.
- They can be difficult to understand and pronounce.
- Scientific naming uses the binomial system. This means it uses two names.
- The genus name begins with a capital letter.
- The whole name is written in italics.

Sometimes a third name is added. For example the dingo is classified as *Canis lupus*, but so is the wolf. A third name is added to show a subspecies, race or variety. For example, the proper naming of the dingo is *Canis lupus dingo*.

☞ **1** Use the notes above to write a paragraph about scientific naming.

_____

_____

_____

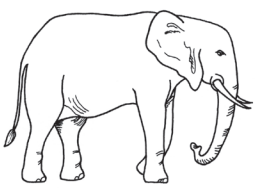
_____

_____

2 Use the names in the word bank to correctly identify the animals shown below. Look carefully at the Latin names for clues.

Word bank

Panthera leo
Felis catus
Canis lupus
Panthera tigris
Proboscidea elephantidae
Canis lupus dingo



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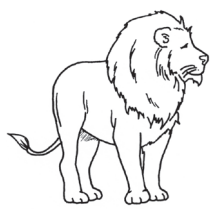

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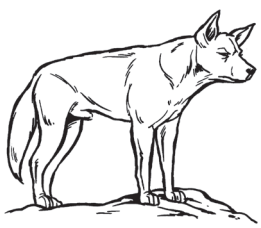

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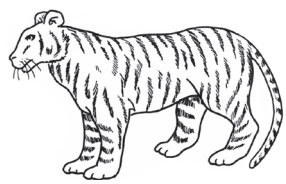

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Dinosaur classification

☞ **1** Use the word bank to complete the following passage.

Dinosaurs are e_____ but they are still classified, just like living animals. Scientists agree that d_____ lived between 250–65 million years ago. They were reptiles that l_____ on land and had straight legs beneath their bodies. The two categories of dinosaurs are known as b_____-hipped and lizard-h_____.

Lizard-hipped dinosaurs are the best known group. They contain all the fearsome m_____-eating dinosaurs that w_____ on three-toed, bird-like feet. They had short a_____ but powerful legs. Also included in this group are the extremely l_____ herbivores. They walked on f_____ feet and had s_____ heads, long necks and long tails.

All of the bird-hipped dinosaurs were h_____. These included the armoured and plated dinosaurs, who had h_____ and spikes. Also in this g_____ are the many two-legged herbivores that travelled in h_____.

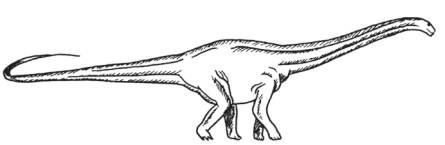
Word bank

herbivores
lived
walked
bird
herds
large
arms
hipped
dinosaurs
extinct
small
four
meat
horns
group

2 Label each dinosaur as bird-hipped or lizard-hipped.

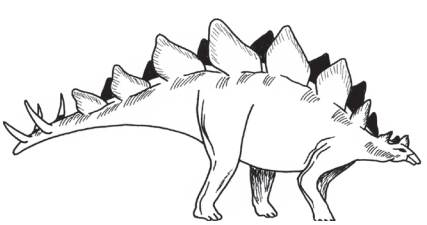
Tyrannosaurus

_____



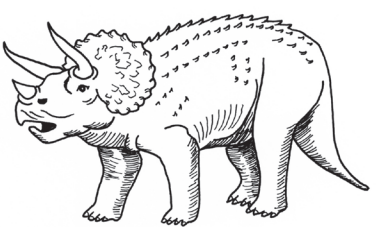
Diplodocus

_____



Stegosaurus

_____



Triceratops

_____

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Teacher's notes – Classification

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Objectives

- To be able to follow a key
- To examine more closely how organisms are classified.
- To identify the five groups of vertebrates
- To determine the three types of mammals

Prior knowledge

Students should be able to sort organisms according to their characteristics. They should be aware that there are different types of vertebrates, insects and plants, and that all of these categories can be broken up into smaller groupings depending on the characteristics.

Background

Scientists need to be able to follow a key to sort organisms into groups. For example, the first step in classifying animals is to determine whether or not an animal has a backbone. Animals are then sorted according to more detailed characteristics, such as: Is it warm blooded? Does it have feathers? Does it lay eggs? Working through these lists of characteristics sorts organisms into order and family groups, where organisms have many characteristics in common with one another.

Starter activity

Working in groups, ask students the difference between a moth and dolphin, then between a moth and a grasshopper, then a seal and a dolphin. Encourage them to list as many characteristics as they can, pointing out any differences. Then narrow the task to a moth and a butterfly, or a whale and a dolphin. Are there still differences?

Activity sheets

The Activity sheet 'Vertebrates and invertebrates' highlights the importance of a backbone in classifying animals. Encourage students to think of as many animals as possible that don't have a backbone. Students can then sort the animals listed into vertebrates and invertebrates and complete the cloze exercise.

The 'Plant classification' Activity sheet introduces the difference between cellular plants and vascular plants before explaining how keys are used to identify organisms. This Activity sheet gives a simplified key for students to complete. Not all plant domains and orders are included, because students would not be familiar with them. Encourage students to think of all the different plants they know and make a class list. See which examples can be slotted into the key.

'Vertebrate classification' explains that there are five groups of vertebrates: fish, reptiles, birds, mammals and amphibians. Students can fill in the table and answer the true/false and 'What am I?' questions to demonstrate an understanding of the characteristics of these five groups.

The Activity sheet 'Mammals' then introduces the three types of mammals: placental, marsupial and monotreme. It may be helpful for students to complete 'The platypus – bird, mammal or reptile?' from Unit 1 before completing this Activity sheet. Students complete a cloze exercise and fill in a table to help clarify the differences between the three types.

The 'Insects' Activity sheet introduces students to arthropod classification, i.e. animals with jointed legs. Encourage students to fill in the table without being intimidated by the Latin/Greek names.

Extension ideas

Devise a key to categorise the students in the classroom. For example, is each student fair- or dark-haired? Are they wearing school shoes or sneakers? See if the class can identify a few groups of students and try to name them.

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Vertebrates and invertebrates

Animals are divided into two groups:

- **vertebrates** (animals with a backbone)
- **invertebrates** (animals without a backbone).

☞ **1** Sort each of the animals listed below into the correct group.

snail worm goat frog crocodile whale human crab spider beetle duck jellyfish tuna dog centipede

Vertebrates	Invertebrates

2 Use the word bank to complete the following passage.

Word bank
vertebrate inside outer backbone invertebrates exoskeleton

A vertebrate is an animal that has a b_____. Its bones are i_____ its body and grow as the animal grows. A dog is a v_____. Animals that don't have a backbone are called i_____. A crab is an example of an invertebrate. It has a hard o_____ shell called an exoskeleton. As a crab grows it sheds its e_____ and grows a new one.

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Plant classification

Plants are divided into **cellular plants** or **vascular plants**. Cellular plants are small, without roots or stems. Vascular plants are plants that have some kind of stem or tube inside to carry water and nutrients to all the parts of the plant.

Word bank

vascular
stems
seed
cones
spores
two
non-flowering
plants
flowering

☞ **1** Use the word bank to fill in the examples of plants listed below.

Mosses are small plants without s_____ or roots.

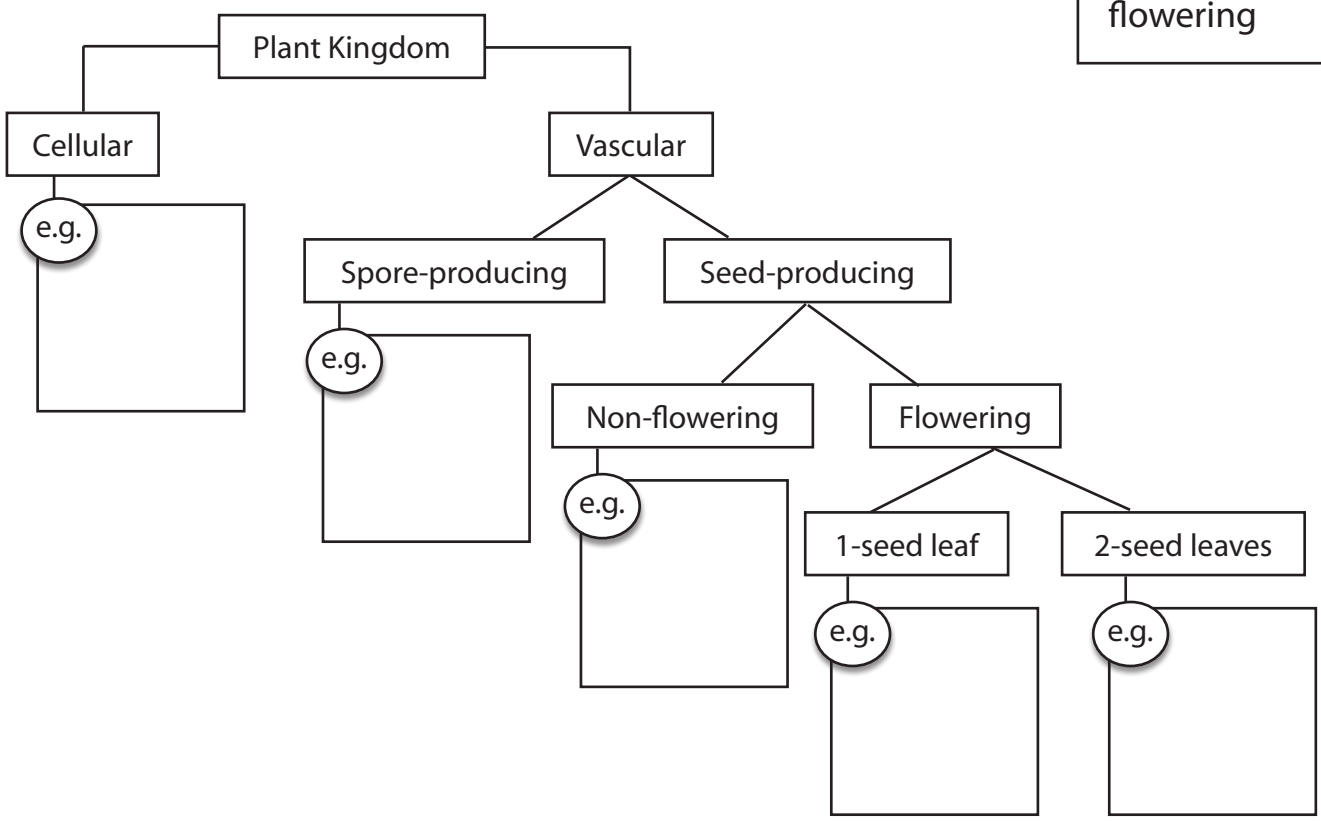
Ferns are v_____ plants that have s_____.

Pines are n_____ plants that have c_____.

Tulips are f_____ plants with one s_____ leaf.

Sunflowers are flowering p_____ with t_____ seed leaves.

2 Use the completed examples to fill in the spaces on the key.



3 Can you think of any other examples to go in any of the boxes above? _____

_____

_____

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Vertebrate classification

Vertebrates can be classified into five different groups. Vertebrates are classified according to how they feed their young, the way they breathe and the type of body covering they have. Mammals and birds are **warm-blooded,** whereas reptiles, fish and amphibians are **cold-blooded**.

☞ **1** Use the word bank to complete the information about these groups in the table below.

Word bank

skin skin and lungs gills lungs feathers lungs lungs fur scales scales

Group	Body covering	Breathes through
amphibians		
birds		
fish		
mammals		
reptiles		

2 Circle 'true' or 'false' for the following statements.

Reptiles have scales. True/False

Mammals have feathers. True/False

Baby birds drink milk. True/False

Amphibians are cold-blooded. True/False

Fish breathe using gills. True/False

Birds are warm-blooded. True/False

What am I?

3 I have wings and I fly at night. I have a furry body and my young drink milk. I am a b_____, which is a type of m_____.

4 I live in the sea and on land in cold climates. I lay eggs. I don't have wings, but I do have flippers. My body is covered with tiny feathers. I am a p_____, which is a type of b_____.

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Mammals

There are three types of mammals: **placental** mammals, **marsupials** and **monotremes**.

☞ **1** Use the word bank to fill out the passage below.

Word bank					
three	milk	eggs	inside	mammal	pouch
mammary	placenta	marsupials	nipple	drinks	grow

Mammals drink m_____ when they are born. The word 'm_____' comes from the term 'mammary gland', which is the tissue that produces milk. There are t_____ types of mammals. Most mammals are placental. They develop i_____ the mother's uterus and are nourished by the p_____, a special organ that the mother produces. When the mammal is born it is well developed and d_____ from the mother's nipples.

When m_____ are born they are tiny, under-developed and skinny. They crawl along the mother's fur until they reach her p_____. They hide inside and attach themselves to the mother's n_____. They drink plenty of milk and g_____ big and strong inside the pouch.

Monotremes develop inside e_____. When the eggs hatch the babies lick milk from the mother's m_____ glands on her belly.

2 Put the following mammals into the correct groups in the table:

koala echidna dog sheep platypus kangaroo

Placental mammals	Marsupials	Monotremes

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Insects

Insects are arthropods, which means they belong to the **Phylum** arthropoda. 'Arthropoda' comes from Greek: *arthron* = joint, + *podus* = foot. There are five **classes** of **arthropods**.

☞ **1** The table below lists the five classes of arthropods. Use the word bank to complete the details in the table.

Word bank

grasshoppers spiders crabs legs body ants antennae water Class

Class	Characteristics	Examples
Class crustacea:	live mostly in w_____	c_____ and lobsters
Class arachnida:	No a_____, eight legs	scorpions and s_____
C_____ diplopoda:	Long round body	millipedes
Class chilopoda:	Long flat b_____	centipedes
Class insecta:	Antennae, three body parts, six l_____	Flies, a_____, bugs, moths, g_____

2 There are many different **Orders** of insects. Classify the following four insects into their correct Order.

butterfly Christmas beetle mosquito stick insect

Coleoptera	Lepidoptera	Phasmatodea	Diptera
Thin hind wings folded under hard front wings. Young are grubs.	Two pairs of wings covered with soft scales. Coiled sucking mouthparts.	Large, long and flat or leaf-like bodies. Slow walking.	One pair of wings only. Sucking mouthparts. Young are like small worms.

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Teacher's notes – Food chains

Achieve!

Objectives

- To know that plants are producers and animals are consumers
- To identify producers and consumers in a food chain
- To understand that consumer populations vary as environmental conditions change
- To recognise the role humans and introduced animals play in disrupting food chains

Prior knowledge

Students must be aware that all organisms need energy to survive. Plants produce their own energy through photosynthesis, but animals must obtain energy from eating.

Background

All organisms need an energy source to function. Plants photosynthesise to produce their own energy (producers), whereas animals need to eat (consumers). Some animals eat plants, others eat animals. For example, the steak we ate for dinner was once a cow that ate grass. Scientists can depict the energy transfer along this process in a food chain.

Starter activity

Ask students to list what they had for breakfast. Collate the data and pick out the four most common meals, e.g. cereal, bacon and eggs. The meals may be very different depending on the background of the students. Establish a food chain for each of the most common meals.

Activity sheets

The Activity sheet 'Food chains' presents some common examples of food chains for students to unjumble. A cloze exercise looks at how heavy metals can be transferred through a marine food chain.

The 'Producers and consumers' Activity sheet presents a complex labelled picture. Students need to list organisms from the picture as either producers or consumers. They then create their own simple food chains based on the organisms in the picture.

The Activity sheet 'Locust and mouse plagues' examines food chains in Australian agricultural systems, looking at locusts and mice and how they depend on our crops. When conditions are good after a severe drought, for example, pest populations can explode. Students complete two food chains demonstrating the effects of such plagues, then analyse a graph of mouse populations over time.

The 'Turtles and pollution' Activity sheet discusses the impact of human activity (pollution) on marine food chains. Students create a food chain, then expand a set of notes to demonstrate their understanding of this issue. Higher order questions at the end of the activity sheet encourage further discussion.

The Activity sheet 'Feral animals – Cats examines the kinds of problems introduced species create in Australian habitats. There are many different feral animals in Australia and all have an impact on native flora and fauna. Students read the paragraph about feral cats and list the main points as a note-taking exercise. They can then answer the true/false statements that follow.

Extension ideas

Students can make a poster depicting a food chain and show how human activity, climatic conditions or introduced species can affect the food chain.

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Food chains

A **food chain** shows how energy is transferred from organism to organism in the food they eat. For example, grain is eaten by a chicken. The chicken is then eaten by a fox. Energy transfers from the grain to the chicken to the fox.

☞ **1** Unjumble the following food chains. Put the first organism to be eaten at the bottom and the last organism at the top.

A cat, caterpillar, lettuce, bird

B grain, fox, owl, mouse

C grass, human, sheep

D snake, fly, frog

E lion, grass, zebra

F fish, shark, sea grass

A	B	C	D	E	F
↑ eaten by	↑ eaten by	↑ eaten by	↑ eaten by	↑ eaten by	↑ eaten by
↑ eaten by	↑ eaten by	↑ eaten by	↑ eaten by	↑ eaten by	↑ eaten by
↑ eaten by	↑ eaten by				

2 Use the word bank to complete the passage below.

Word bank

fish sea eat bigger plankton livers poison metals stored food lots

Sometimes poisonous metals are washed into the s_____. The quantities are very small and the metals are absorbed by the p_____ in the sea. Small f_____ eat the plankton and are not harmed by the metals. The poison is s_____ in their livers. Bigger fish such as tuna e_____ lots of the small fish. Each time these b_____ fish eat the small fish they eat the stored p_____ too. The bigger fish have bigger l_____ and can tolerate more poison. A shark then eats l_____ of the bigger fish and takes in the poison too. This is how poisonous m_____ can accumulate in a f_____ chain.

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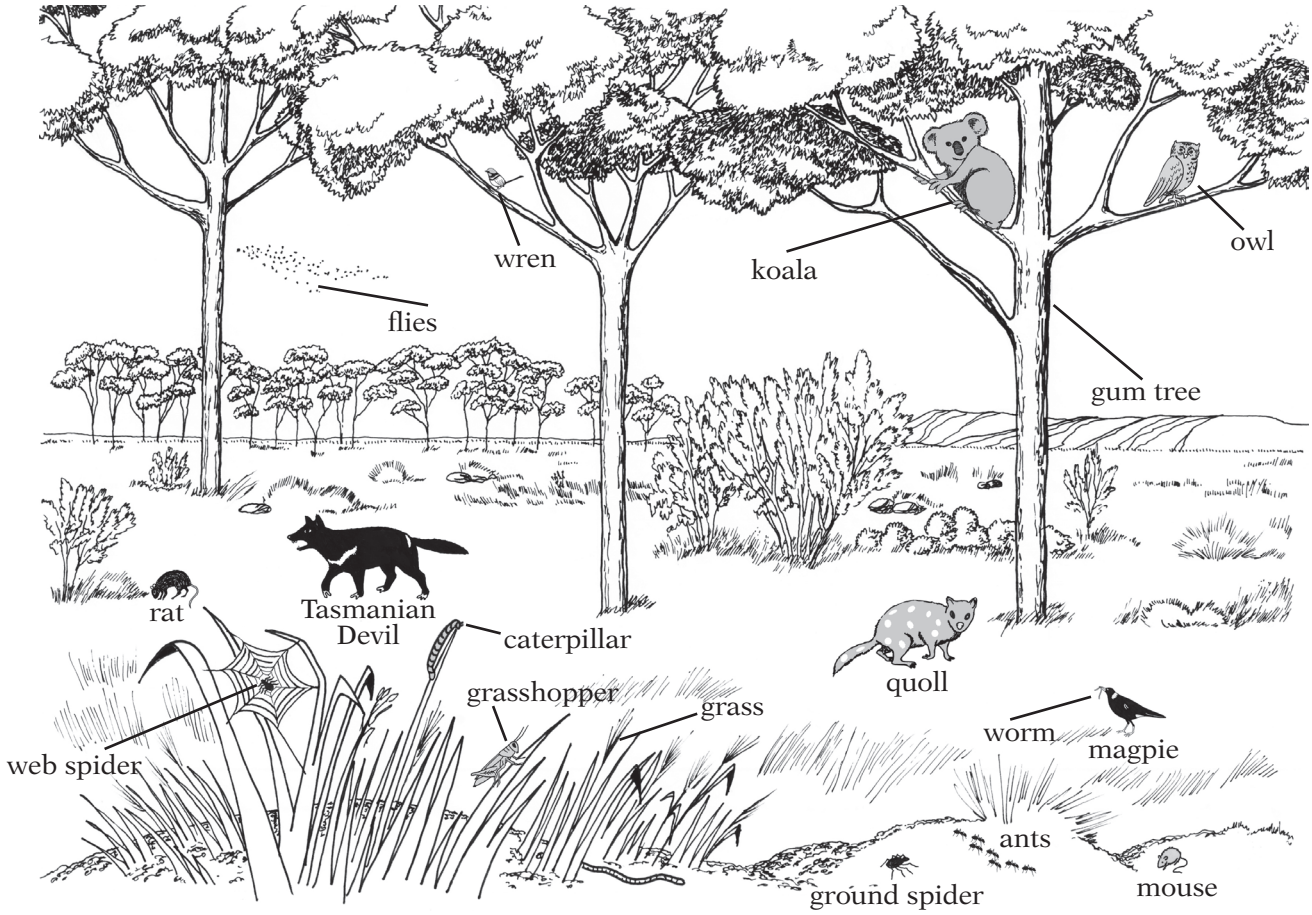
Producers and consumers

All living things need an **energy source**. Some living things, such as plants, produce their own energy. They are **producers**. Others need to eat plants or animals to gain energy – they are **consumers**.



☞ **1** Study the picture above. Circle the **producers** in blue and the **consumers** in red.

2 List three possible food chains found in the picture above.

consumers			
	↑ eaten by	↑ eaten by	↑ eaten by
consumers			
	↑ eaten by	↑ eaten by	↑ eaten by
producers			

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Locust and mouse plagues

Mice and locusts are common **pests** on Australian farms. They take advantage of the extra food that farms provide, eating the pasture, crops and grains. Both animals can breed very quickly and reach **plague numbers** within months if conditions are favourable.

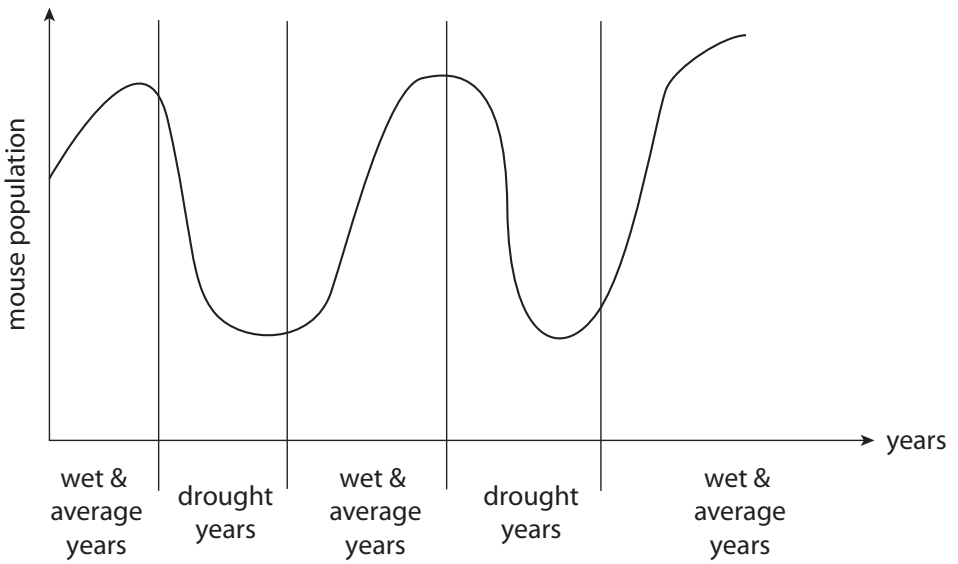
☞ **1** Use the word bank to complete the food chains for these two Australian pests.

s_____	b_____
↑ eaten by	↑ eaten by
m_____	l_____
↑ eaten by	↑ eaten by
g_____ and g_____	any green s_____

Word bank

grass
mouse
grain
snake
bird
locust
shoots

Study the graph below, then answer the questions.



2 In drought years, mouse numbers _____.

3 In wet years after drought, mouse numbers _____.

4 Why do the numbers rise so quickly? _____

_____

5 Predict what might happen to bird and snake numbers during mouse or locust plagues.

_____

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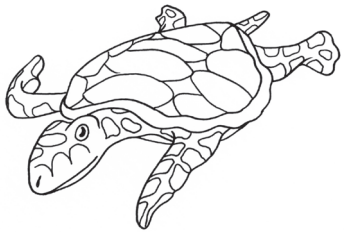
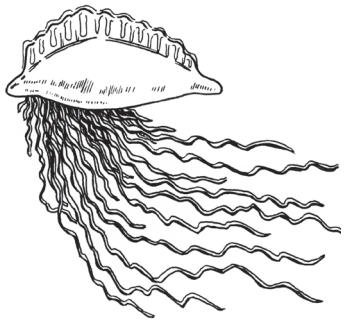
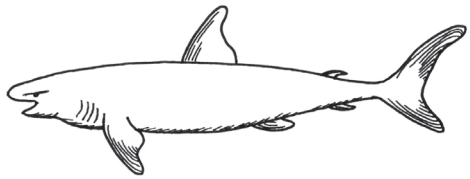
Turtles and pollution



sea turtle



jellyfish



shark

☞ **1** Make a food chain using the three animals in the picture above.

↑ eaten by
↑ eaten by

- Turtles eat jellyfish.
- People throw plastic bags into the sea.
- Plastic bags block the intestine.
- Bag with air inside makes diving difficult.
- Plastic bags look like jellyfish.
- Dead turtle decomposes quickly.
- Plastic can take years to break down.

☞ **2** Use the notes above to write a paragraph about the dangers of plastic bags in our oceans.

_____

_____

_____

_____

3 Why is it important for turtles to dive? _____

4 If a turtle with a plastic bag lodged in its digestive system dies in the ocean, what will happen to the plastic bag? _____

_____

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Feral animals – Cats

Feral cats are found in all areas of Australia. They spend most of the day hiding and sleeping, and hunt at night. Feral cats don't drink much water, as they get moisture from the animals they kill. They kill birds, small mammals, reptiles and frogs. Some larger cats will even kill animals the size of a possum. Feral cats hunt many **native species** and ground-nesting birds. Scientists believe they have hunted many species of small desert animals to **extinction**. Bilby, numbat and mala populations have all been seriously affected by the feral cat population.

Feral cats breed throughout the year and have few **predators**, except dingoes and eagles. They are hard to control because they don't take baits and are shy of traps. They avoid humans and so are hard to shoot.

☞ **1** In your own words, write four important ideas about feral cats.

_____

_____

_____

_____

2 Are these statements true or false?

a) Feral cats sleep during the night. _____

b) Feral cats only eat small mammals. _____

c) Bilby populations are threatened by feral cats. _____

d) Feral cats are producers. _____

e) Feral cats are at the bottom of the food chain. _____

3 List any other feral animals you can think of.

_____

_____

_____

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Teacher's notes – Food webs

Achieve!

Objectives

- To identify links between organisms in a community
- To recognise the importance of microorganisms as consumers/decomposers
- To understand the difference between aerobic and anaerobic decomposition
- To predict what changes could occur in a community if a new species is introduced

Prior knowledge

Students should have an understanding of food chains and the concepts of producers and consumers.

Background

Food chains don't exist in isolation to one another. Many plants are eaten by a vast number of creatures. Many organisms eat a range of foods within their habitat. Overlap occurs and relationships are complex. Microorganisms, fungi and insects all play a role, often as decomposers, in a food web.

Starter activity

Revise food chains with students and then ask, 'Who eats the consumer at the top of the chain?' Brainstorm a list of decomposers, then link these back into the food chain, constructing a complex food web.

Activity sheets

The Activity sheet 'Food webs' demonstrates the way a food web shows links between organisms in a habitat. Students are asked to draw their own food web of well-known savannah species.

The 'Microorganisms as consumers' Activity sheet explains the importance of bacteria and fungi as decomposers. Students complete a cloze activity to explore the topic. Then, using a picture of a dead mouse as an example, they label where the various decomposers act on the carcass.

'Decomposition in action' presents the steps involved in composting, which students need to put in the correct order. A graph of temperatures inside a compost heap illustrates temperature fluctuations. Encourage students to explain why these temperature changes occur.

The Activity sheet 'Introduced species – Cane toads' presents jumbled sentences about cane toads – a serious pest in Australia. Students rewrite the sentences in full, then draw a food web demonstrating the role of the cane toad within it.

The 'Introduced species – Rabbits' Activity sheet explores how rabbits damage the Australian environment. Students complete a cloze passage, then use the information to draw up a food web illustrating how rabbits can impact on other species. Students then examine the factors that make rabbits such effective invaders.

The Activity sheet 'Introduced species – Lantana' then builds on this idea, demonstrating that plants can also be invasive species. Students complete a cloze passage and answer questions to demonstrate their understanding that lantana is a serious introduced weed.

Extension ideas

Adapt the game of 'celebrity hats' to 'celebrity pest'. Challenge the students in the hot seat to ask questions until they can guess which pest they are, based on the answers they receive from the audience.

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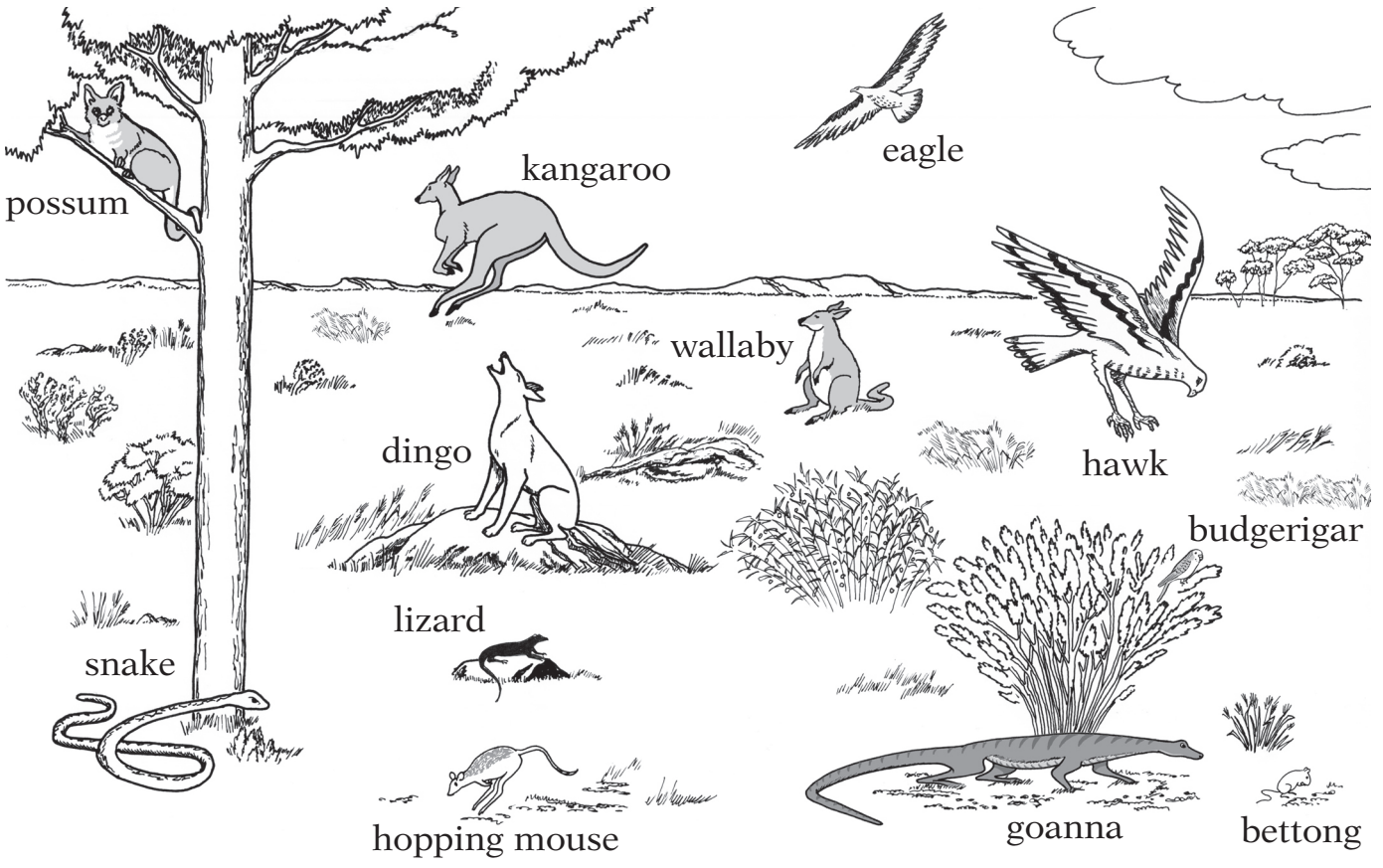
Achieve!

Food webs

A **food web** shows the links between all the organisms that live in a particular **habitat**. A food web can include lots of different individual food chains.

☞ **1** Draw links between the organisms in the picture to show who eats what.



2 In your workbook, draw your own food web using the following savannah animals and plants.

zebra	wildebeest	giraffe	gazelle	lion	cheetah
vulture	dung beetle	snake	hyena	jackal	meerkat
warthog	mouse	baboon	fly	centipede	grass
shrubs	trees	hare			

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Microorganisms as consumers

☞ **1** Use the word bank to complete the passage below.

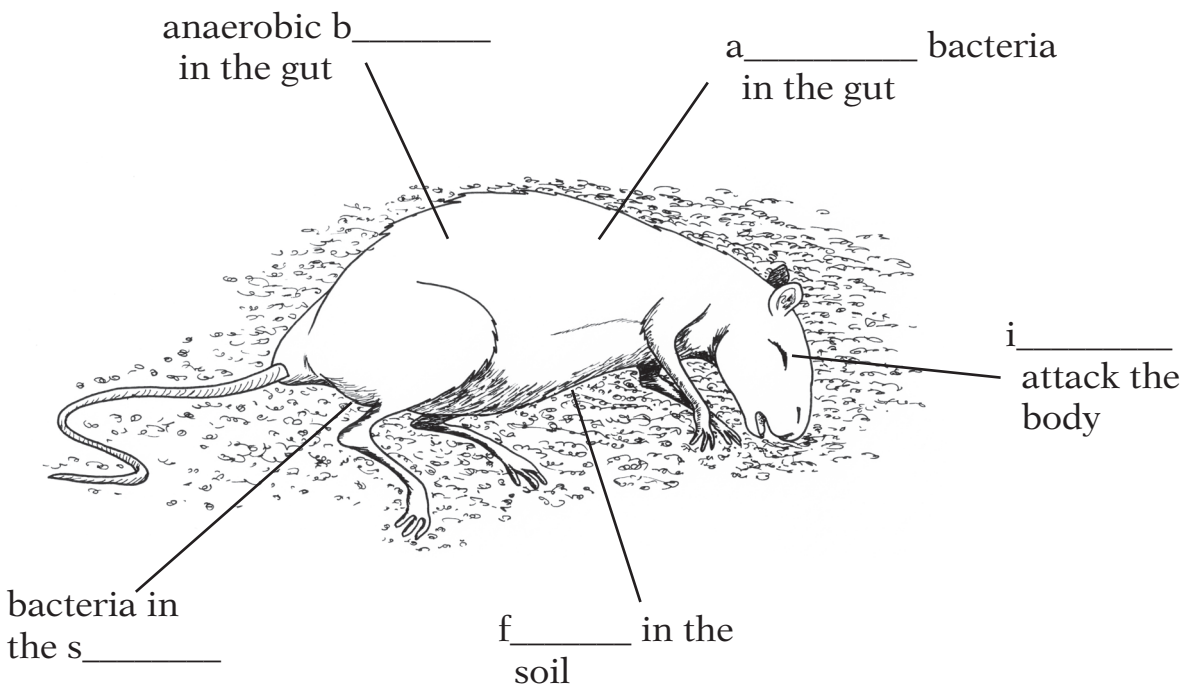
Word bank					
flies	gas	fungi	microorganisms	oxygen	Ants
bloat	rotting	**aerobic**	bacteria	decomposition	carcass

When an animal dies, m_____ (mainly bacteria and fungi) start consuming the carcass. Firstly, a_____ bacteria inside the animal start to feed and multiply using what o_____ is available. Their activity produces lots of g_____, making the carcass expand, or b_____. When the available oxygen is used up, **anaerobic** b_____ take over. Bacteria and f_____ in the air and soil also start to consume the c_____.

Insects such as ants and f_____ also take part in d_____. A_____ eat the dead tissue. Flies lay their eggs in the r_____ flesh, which the young maggots eat.

2 Use these words to complete the labels on the diagram.

insects	soil	bacteria	fungi	aerobic



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Decomposition in action

☞ **1** Put the steps below in the correct order to show how to make a compost heap.

- Water the heap to keep it moist.
- Collect manure, food scraps, leaves, soil, straw and shredded paper.
- Set aside an area for the heap.
- Cover the heap.
- Turn the heap once each week.
- Lay down manure, scraps, straw, soil, leaves and paper in layers.

1

↓

2

↓

3

↓

4

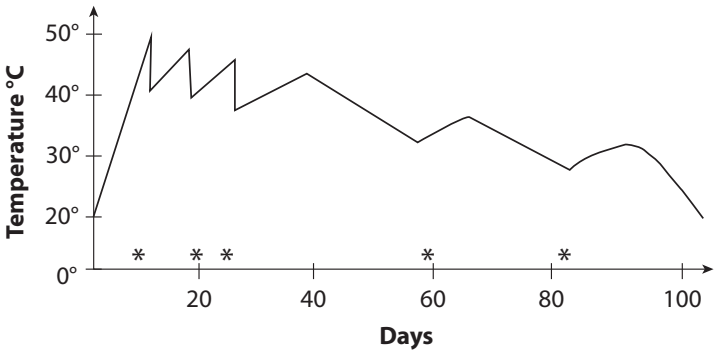
↓

5

↓

6

Compost forms best if it is moist and has plenty of air. Very wet or very dry compost heaps break down slowly. The graph below shows the temperature in an uncovered compost heap.



* days when the heap was turned

☞ **2** Why does the temperature drop over time? _____

_____

3 What happens when the heap is turned? _____

_____

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Introduced species – Cane toads

☞ **1** Unjumble the sentence parts and then rewrite the information about cane toads in the correct order.

a temperature range of Can survive in 5°C – 40°C.

_____

in warm Active at weather. night

_____

crevices during Shelter in the day. moist

_____

eggs at a time. lay 8000–35 000 Females

_____

brackish (salty) Tadpoles water. can tolerate

_____

breed within Grow quickly and a year.

_____

throughout its Poisonous life cycle.

_____

in Australia. very few predators Has

_____

tadpoles, frogs and small mammals, Eats insects, dead creatures.

_____

2 In your workbook, draw a food web linking the following animals.

goanna	snake	frog	cane toad	flies
beetles	dingo	marsupial mouse		

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Introduced species – Rabbits

☞ **1** Use the word bank to complete the passage below.

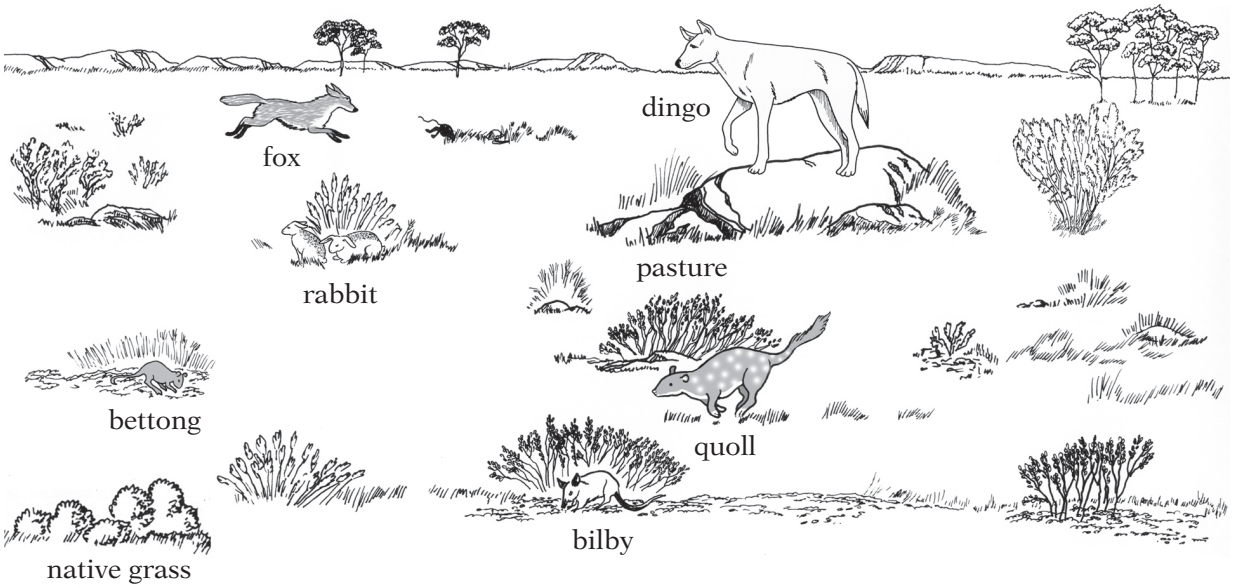
Word bank

numbers ships sea species female quoll bettong transport quickly
wild burrowed compete widespread rabbits ate animals recover

European settlers brought rabbits to Australia in their s_____. They bred q_____, were easy to t_____ and provided meat for the long months at s_____. Over the next hundred years r_____ were released into the w_____ in Australia. Their n_____ swelled because **predators** like the q_____, the dingo and introduced foxes were not w_____ or abundant.

Rabbits a_____ the new pastures that farmers grew. They also ate native grasses and b_____ into the soil. There is even evidence that f_____ rabbits have taken over the burrows of native a_____. Rabbits c_____ with native burrowing animals such as the b_____ and the bilby for food. In times of drought, numbers of all these s_____ drop, but when times are good, rabbit numbers r_____ more quickly than marsupials.

2 Draw arrows and produce a food web to show the relationships between the animals and plants in the picture.



3 List in point form the factors that make the rabbit a successful coloniser.

- _____
- _____
- _____
- _____

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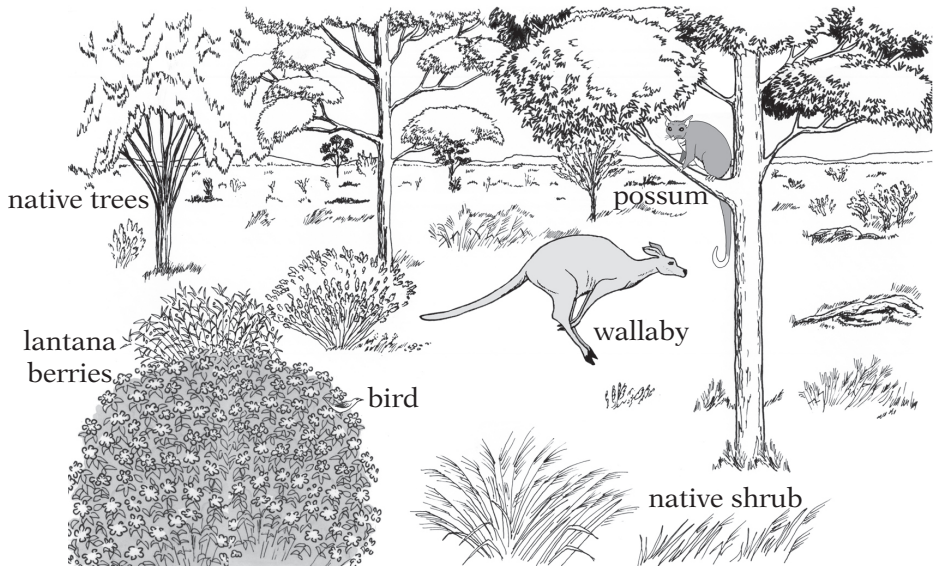
Introduced species – Lantana

☞ **1** Use the word bank to complete the passage below.

Word bank
eat plant poisonous weed soil stems Seeds native spread grows riverbanks flowers

Lantana is a plant **native** to Mexico and Central America. After it was introduced to Australia it s_____ rapidly, and has become a serious w_____ problem. Lantana thrives in cleared areas or sunny spots along r_____. The leaves are p_____ to animals. Because nothing will eat lantana, it g_____ quickly, forming a dense thicket with prickled s_____. It smothers n_____ vegetation and also releases a chemical into the s_____ that prevents **germination** of other plants. Lantana f_____ are colourful and attractive. Birds e_____ the berries. S_____ are then spread in the birds' droppings. One lantana p_____ can produce up to 12 000 seeds per year.



2 Draw arrows linking the organisms in the illustration to create a food web showing what foods they eat.

3 As the lantana population grows in a habitat, what will happen to the native trees and shrubs? _____

4 What will happen to the possums and wallabies over time? _____

_____

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Teacher's notes – Looking at cells

Achieve!

Objectives

- To be able to identify parts of a light microscope
- To be able to operate a microscope correctly and safely
- To understand that different microscopes are used for looking at different specimens
- To be able to follow the steps in preparing a slide
- To recognise the differences between plant, animal and fungi cells

Prior knowledge

Students should be aware of germs, viruses and bacteria – things that are too small to see with the naked eye. They should be able to identify basic differences between plants and animals and fungi.

Background

Only a few hundred years ago scientists were unaware of cells, which are the building blocks of all living things. With the invention of the light microscope scientists were able to examine plant and animal tissue, identifying things far smaller than the naked eye could see. The electron microscope was invented in the nuclear age and has made incredible magnification possible – in some instances to the molecular level.

Starter activity

Ask students to examine a detailed drawing, such as a five dollar note. Ask them to identify how many windows are on the drawing of Parliament House, or a similarly minute detail. How could they see small things better? Give students magnifying glasses to better study the detail. What about the surface of their skin? What details can they see with the magnifying glass?

Activity sheets

The Activity sheet 'Light microscope' provides a diagram of a light microscope for students to label, to help familiarise them with what each part does. Encourage the class to list possible rules of microscope use and safety.

The 'Electron microscope' Activity sheet introduces electron microscopes and explains that they are far more powerful than light microscopes. Students answer questions to explore how electron microscopes work and consider their possible uses.

In 'Slide preparation' students order the steps involved in preparing an onion slide. They then prepare their own slide, and draw two pictures of it at different magnifications. Students then match statements about slide preparation to consolidate what they have learnt.

The 'Name that cell' Activity sheet revises students' knowledge of the differences between animal, plant and fungi cells. Students answer 'What am I?' and true/false questions then complete a table.

Extension ideas

Ask students to look online or in magazines to find images of objects or animals that have been examined with an electron microscope. Make a collection of these pictures and display them in the classroom.

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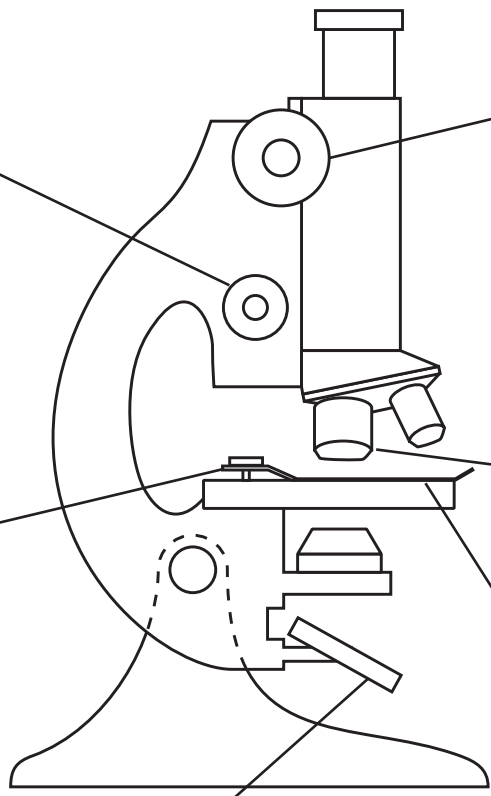
Light microscope

☞ **1** Use the word bank to finish labelling the parts of the microscope.

Word bank

look focus clips mirror fine focus knob light slide stage

objective lens eyepiece object focus knob carefully put



The e_____ is the part you l_____ through.

The f_____ f_____ k_____ adjusts the focus c_____.

The f_____ k_____ adjusts the f_____.

The o_____ l____ magnifies the o_____ on the slide.

The c_____ hold the s_____ in place.

The m_____ reflects l_____ to highlight the slide.

The s_____ is where you p___ the slide.

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Electron microscope

Electron microscopes

- Electron microscopes are far more powerful than light microscopes.
- They were invented in the 1930s.
- Electron microscopes use electrons that are sped up in a vacuum.
- Beams of electrons are directed at the specimen.
- Electrons are either absorbed or scattered to form an image on an electron-sensitive photographic plate.
- Electron microscopes are large, heavy and expensive.
- The microscope magnifies microorganisms, cells, crystals, metals and large molecules.

Read the information above and then answer the following questions.

☞ **1** In what way could electron microscopes be useful in medical research?

_____

_____

2 Living cells can't survive in a high vacuum. How is this a problem when studying cells using an electron microscope? _____

_____

3 List reasons why schools don't have electron microscopes. _____

_____

4 How might palaeontologists (scientists who study fossils) find this microscope useful?

_____

_____

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Slide preparation

☞ **1** Ruby is going to prepare a slide of an onion skin. Number the steps she needs to take in the correct order.

___ Place the cover slip over the specimen.

___ Take a very thin slice of the onion specimen.

___ Add stain with a dropper.

___ Collect the slide, cover slip, droppers, water and stain.

___ Add a drop of water.

___ Place the specimen on the slide.

___ Blot up excess water and stain.

2 Follow Ruby's instructions to examine your own onion slide. Then draw the specimen you have studied under a microscope in two different magnifications. Label the drawings clearly.

3 Match the statements below.

The cover slip stops the lens	at an angle to prevent air bubbles.
Adding a stain	flat; others have a well.
Some slides are	from touching the specimen.
Lower the cover slip	makes the cells easier to see.

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Name that cell

☞ **1 What am I?**

I have several nuclei. I have a cell wall made of chitin but I don't contain chloroplasts. I grow at the ends. I am a _____ cell.

I have no chloroplasts and no cell wall. I am a round sort of shape and I can move. I am an _____ cell.

I have a rigid cell wall and I contain chloroplasts. I have a vacuole and a nucleus. I am a _____ cell.

2 Label these statements 'true' or 'false'.

a) Animal cells have a cell membrane. _____

b) Fungal cells have just one nucleus. _____

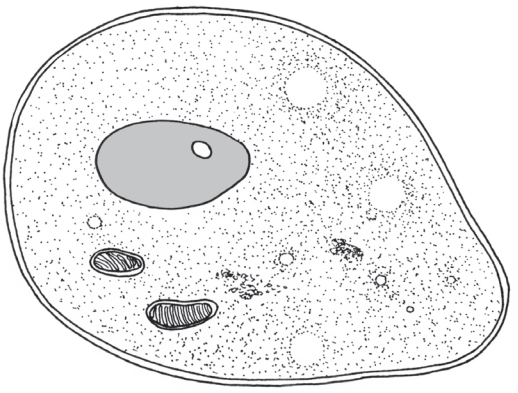
c) Animal cells have chloroplasts. _____

d) Plant and fungal cells have a vacuole. _____

e) Only animal cells have mitochondria. _____

3 Use 'yes' and 'no' to complete the information in the table below.

	Animal cell	Plant cell	Fungal cell
Cell wall			
Cell membrane			
Vacuole			
Movement			
Chloroplasts			



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Teacher's notes – Cell structures

Achieve!

Objectives

- To recognise that some organisms are single-celled
- To understand that cells make up specialised tissues in complex organisms
- To be able to describe the four stages of cell division
- To recognise that cell division and tissue culture can be used by industry

Prior knowledge

Students should be able to identify plant, animal and fungi cells. They should also be familiar with cell bodies such as the nucleus, vacuole and mitochondria.

Background

Cells are the building blocks of life but they are not all alike. Some cells are single-celled, or uni-cellular, organisms. These are generally microscopic, although yeast is a well-known exception. However, complex organisms that we can see with the naked eye are made up of many different groups of specialised cells. Specialised cells form together to create specialised tissues, even within something as small as a leaf. Similarly, our bodies have many types of specialised cells that have different functions.

Starter activity

Have students work in groups to think of a house and the different materials it is made from. Do all houses have a tin roof? Are all floors made of timber? Are all houses clad in brick? What is the purpose of all these different materials? (To keep the rain off, keep the heat in, etc.) One brick isn't sufficient to build a house, but many bricks can. What about the windows and doors? Why aren't they made of bricks? Start thinking about cells and how they are different and form specialised tissues.

Activity sheets

The Activity sheet 'Animal cells and tissues' presents a labelling exercise on animal cells. Students then read a list of notes on blood cells and summarise the information in their workbooks.

The 'Plant cells and tissues' Activity sheet explores the way that plant leaf cells show a range of specialisation according to the job each cell is required to do. Students use this information to label a leaf diagram, then complete a cloze passage about the tissues in a typical leaf, including chloroplasts.

Microorganisms don't all look alike. The 'Single-celled organisms' Activity sheet examines protozoa and the mode of movement each deploys. Students summarise the key ideas in a passage introducing ciliates, flagellates and amoebas, then examine and identify the pictures. The lesson concludes with some true/false questions to consolidate students' understanding.

'Cell reproduction in animal cells' presents the four stages of mitosis in table form. Students need to complete the table using the word bank provided, then answer some questions.

The Activity sheet 'Tissue culture' explains that many commercial plants are now produced using tissue culture. This work sheet presents notes about tissue culture using plant materials. Students can expand the notes to write a paragraph on the topic.

Extension ideas

Ask students to design their own alien life form to present to the class. It should have several groups of specialised tissues that need to be explained.

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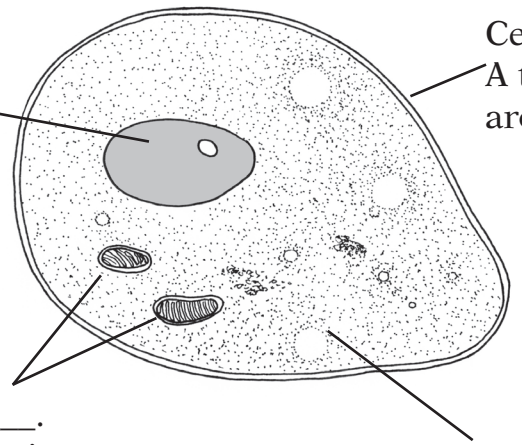
Animal cells and tissues

☞ **1** Use the word bank to label the diagram of an animal cell.

Word bank

Nucleus membrane gas control thin Cytoplasm substance Mitochondria



N_____. Contains chromosomes. Is the c_____ centre of the cell.

Cell m_____. A t_____ skin around the cell.

M_____. Plays a part in g____ exchange.

C_____. Jelly-like s_____ where processes occur.

2 Use the notes below to write a paragraph about blood cells.

Blood in mammals

- Blood is 90% water.
- There are millions of red blood cells per drop.
- Red blood cells remove waste and deliver oxygen.
- Red blood cells don't have a nucleus.
- There are thousands of white blood cells per drop.
- White blood cells attack germs.

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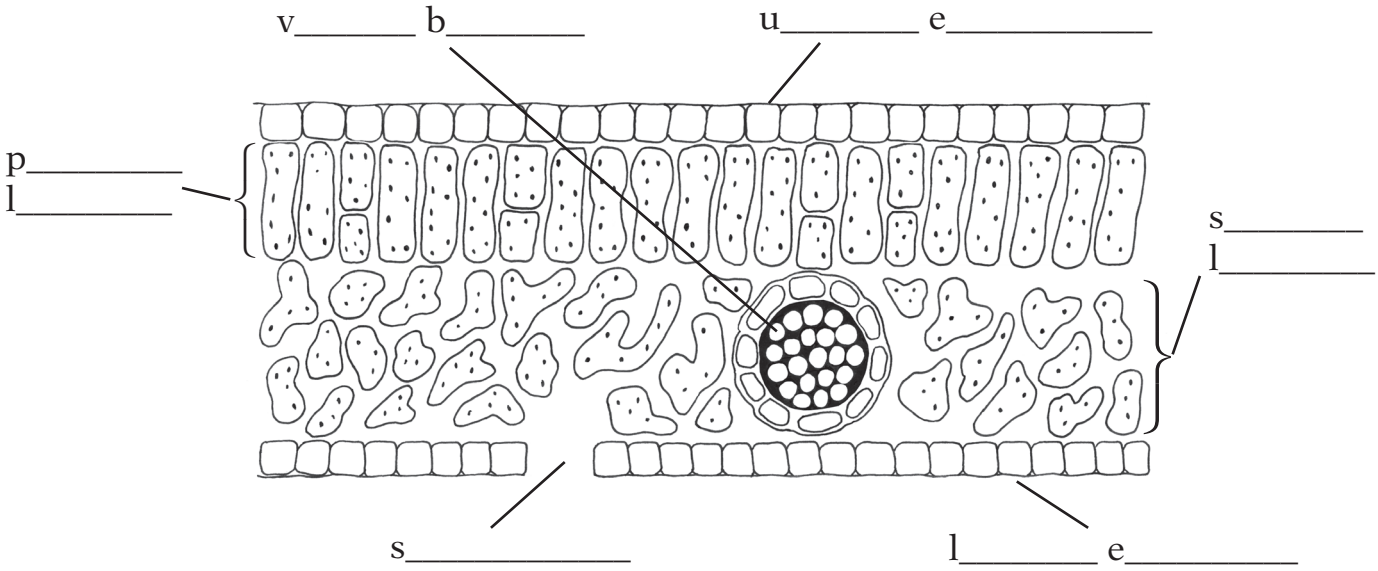
Plant cells and tissues

☞ **1** Label the diagram of a plant cell below using the words provided.

lower epidermis	stomata	upper epidermis
vascular bundle	spongy layer	palisade layer



2 Use the word bank to complete the passage below.

Word bank

carbon palisade leaf layer small oxygen chloroplasts stomata photosynthesis epidermis vascular cells upper sunlight

The lower e_____ is tissue formed by cells on the underside of the l_____. It has many s_____ holes in it, called **stomata**. The s_____ allow oxygen and c_____ dioxide to flow in and out of the leaf. In p_____, carbon dioxide enters through the stomata and is exchanged for o_____ in the cells of the spongy l_____. These cells contain a few c_____. The p_____ layer is tissue which contains most of the **chloroplasts**. It is just below the u_____ epidermis, which is the thin upper layer of the leaf. It is thin to allow s_____ to penetrate through to the palisade layer. The v_____ bundle is the group of c_____ forming vein tissue within the leaf.

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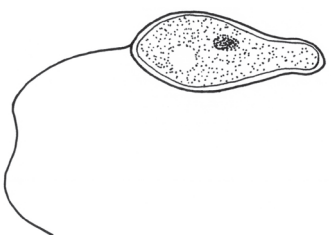
Single-celled organisms

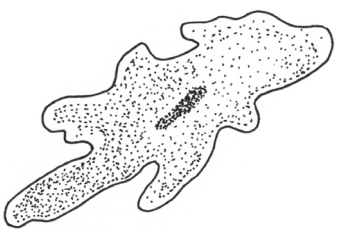
☞ **1** Read the passage below and list the four main points.

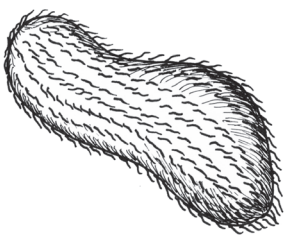
If you take a drop of pond water and inspect it under a microscope, you will see some small shapes. Some are green and are probably single-celled plants. Some are colourless and move about eating other shapes. These are very simple **single-celled** animals called **protozoa**. Protozoa are classified into groups according to how they move. **Ciliates** have many small hairs that move in waves, making the cell move very fast. **Flagellates** have one or more long whip-like structures that thrash about, propelling the cell along. **Amoebas** move in a flowing motion, their shape changing from moment to moment. Their body substance flows into slender, finger-like extensions as they move.

- _____
- _____
- _____
- _____

2 Identify the three types of protozoa drawn below.







_____ _____ _____

3 Write whether you think each of these statements is true or false.

Statement	True/False
Amoeba are single-celled plants.	
Ciliates move by waving lots of small hairs.	
Flagellates have a whip to attack other protozoa.	
All single-celled organisms are animals.	
Protozoa are single-celled animals.	
We need a microscope to see many protozoa.	

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Cell reproduction in animal cells

☞ **1** Use the word bank to fill in the table below.

Word bank

four	two	**anaphase**	chromosomes	thicken
sets	three	fibres	visible	spindle
stage	begins	disappears	separates	membranes

Stage	Name	Description
one	Division b_____ '**prophase**'	Centrioles become v_____, separate and move to opposite ends of the cell. Chromosomes become visible, shorten and t_____. Nuclear membrane d_____.
two	Middle s_____ of nuclear division '**metaphase**'	Fibres extend from each centriole forming a s_____, an oval shape. Chromosomes line up, lying along the spindle f_____.
t____	Separation of chromosome material '**a**_____'	Chromosome material s_____ forming two similar sets of c_____ that move towards the spindle ends.
f_____	End of mitosis '**telophase**'	Two s_____ of chromosomes move further apart. Nuclear m_____ form around each set of chromosomes, making t____ new cells.

2 How many phases are there of cell reproduction? _____

_____

3 What happens to the nuclear membrane during cell reproduction? _____

_____

4 What important material is reproduced in each cell? _____

_____

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Achieve!

Tissue culture

Plant tissue culture

- Plants are grown in laboratories from cells.
- This process is called tissue culture.
- The mother plant is called the 'explant'.
- Plants grown in mineral solution.
- Plants are grown in disease-free conditions.
- Every daughter plant is a clone of the parent explant.
- Used for flower and food industries.

☞ **1** Use the notes above to write a paragraph about plant tissue culture.

_____

_____

_____

_____

_____

2 Match the statements below.

Animal and plant cells	sterile to prevent disease.
The conditions are	in the growth medium.
Tissues grow rapidly	can be grown in laboratories.

3 Some laboratories grow animal and human cells. Why would they do this? _____

_____

_____

4 How has modern society benefited from growing cells in laboratories? _____

_____

_____

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Teacher's notes – Energy systems

Achieve!

Objectives

- To know the different parts of the digestive system
- To understand the function of parts of the digestive system
- To be aware of different digestive systems in different animals
- To know that photosynthesis is dependent on sunlight
- To be able to test for oxygen in a photosynthesis practical exercise

Prior knowledge

Students should be aware of food chains and how our bodies use food to provide us with energy. They should be able to name some parts of the digestive system, such as the stomach and intestine. They should also be able to identify plants as producers.

Background

Green plants use water and carbon dioxide in the presence of sunlight to produce glucose and oxygen. This process is called photosynthesis and is the way green plants produce their own energy. Consumers need to eat plants to obtain energy. Nutrients and glucose are derived from food via the digestive system. Humans have a straightforward digestive system, but other animals have more complex digestive systems.

Starter activity

Working in groups, ask students to list all the parts of the digestive system. Ask them to draw a diagram. (Make sure to remove any classroom posters or textbooks they could copy from.) Have each group present their diagram to the class, and ask the audience to give feedback. Draw up a detailed accurate diagram from the class activity. Discuss whether all mammals have a similar digestive system.

Activity sheets

The Activity sheet 'Human digestion' presents a crossword puzzle introducing terms relating to human digestion. Students also label a diagram of the human digestive system.

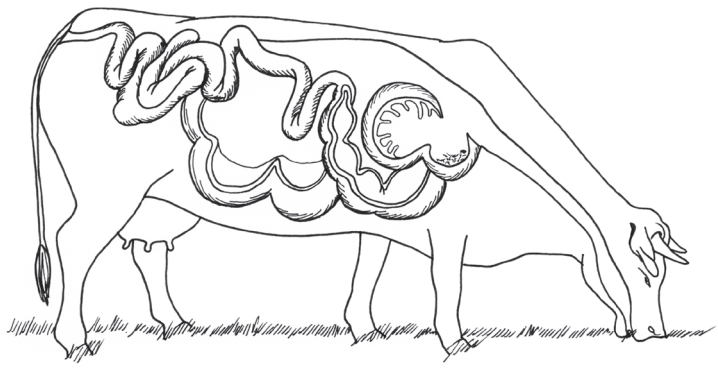
The 'Human vs cow digestion' Activity sheet explains that animals that chew the cud are called ruminants. Simplified diagrams of ruminant (e.g. cow) and monogastric (e.g. human) digestive systems are provided for students to label. There are higher order questions to complete and a true/false activity.

The Activity sheet 'Sea anemone vs bird digestion' explains that sea anemones have a very basic digestive system compared to that of a bird. Students complete a cloze exercise on sea anemones and label a diagram of a chicken's digestive system, then consolidate the information by responding to some true/false statements.

'Photosynthesis' presents steps in a common laboratory test. Students need to order the steps in the practical exercise, then conduct the experiment and answer the questions provided about the by-product of photosynthesis. Students will each need a toothpick, test tube, matches and a beaker in order to perform this experiment. Supervision will be needed to ensure student safety.

Extension ideas

Ask students to research the digestive system of a non-mammal. Examples could be a mollusc, fish, bird, worm or octopus.



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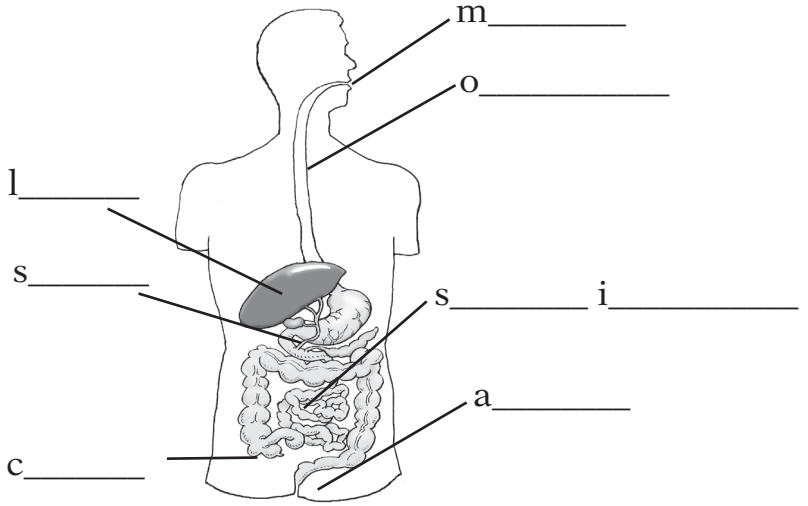
Human digestion

☞ **1** Label the diagram below using the words from the word bank.

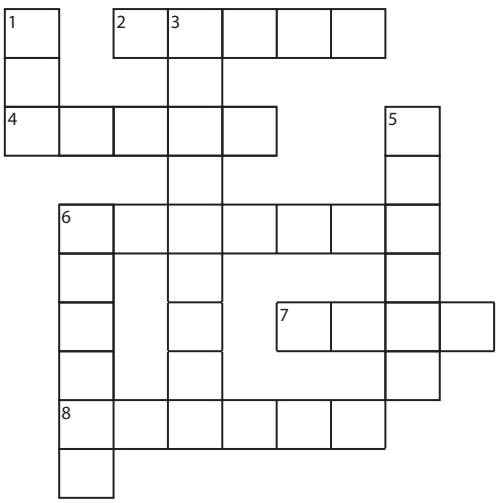


Word bank

stomach
colon
small intestine
mouth
anus
liver
oesophagus

2 Complete the crossword puzzle using the words below.

teeth	energy	eat	gullet	gastric	liver	rectum	intestine	anus



Across

2. Organ that makes bile
4. _____ chew the food into small pieces
6. _____ juices digest food
7. Opening of the rectum
8. Food is needed for _____

Down

1. Humans _____ food
3. The small _____ is where most nutrients are absorbed
5. Faeces passes into the _____ then out the anus
6. Oesophagus

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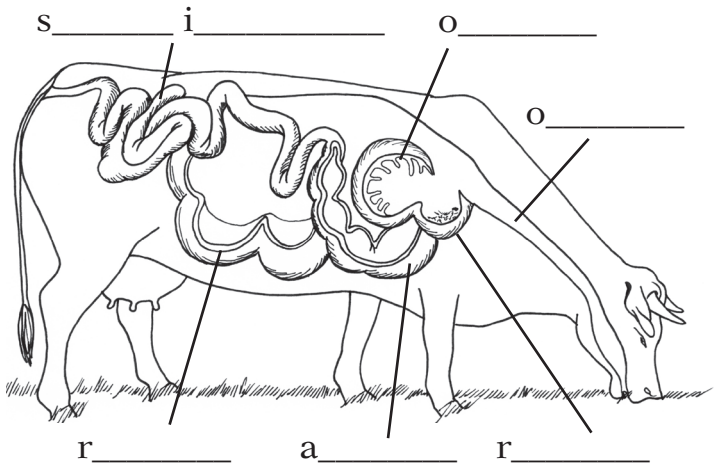
Human vs cow digestion

Human digestive systems are called **monogastric** systems. A cow's digestive system is called a **ruminant** system. Ruminants have four stomachs, beginning with the rumen, then the reticulum, the omasum and finally the abomasum. Chewed grass stored in the rumen is regurgitated back into the mouth and the cow chews it again. This is called chewing the cud. The cud is then swallowed and passes to the reticulum.

☞ **1** Use the word bank to label the diagram of the cow's digestive system.



ruminant system

Word bank

oesophagus
small intestine
rumen
reticulum
abomasum
omasum

2 Can you list two other animals that 'chew the cud'? _____

3 Why do ruminants 'chew the cud'? _____

_____

4 If humans ate grass, what would happen to the grass during digestion? _____

_____

5 Label these statements true or false.

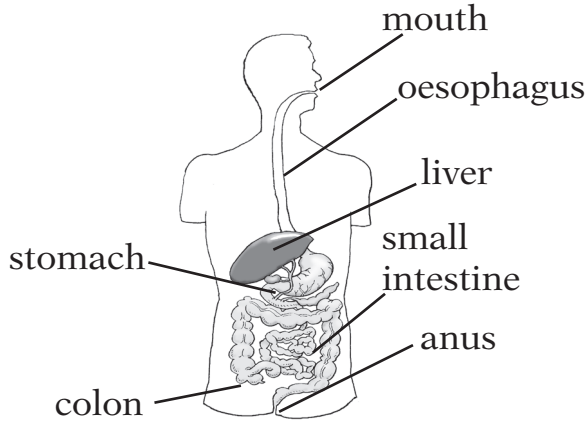
a) Cows chew the cud. _____

b) Cows have three stomachs. _____

c) The rumen is the third stomach. _____

d) Humans only have one stomach. _____

e) Both cows and humans have an intestine. _____



monogastric system

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Sea anemone vs bird digestion

Sea anemones have a simple digestive system.

Word bank

prey
animals
digested
ejected
brushes
poisonous
mouth
bodies
paralysing

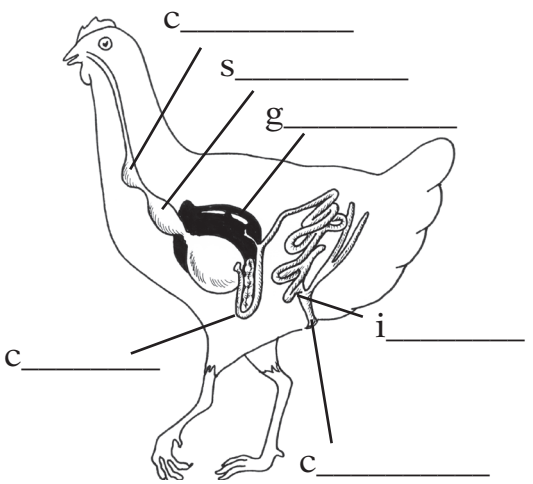
☞ **1** Use the word bank to complete the passage below.

Sea anemones are predatory a_____ with poisonous tentacles. They have column-shaped b_____ with an adhesive foot at one end and a m_____ at the other. The mouth is surrounded by p_____ tentacles. When a fish b_____ past, the tentacles shoot poison, p_____ the fish so it can't move. The tentacles bring the p_____ into the sea anemone's mouth. The prey is d_____ inside the digestive cavity within the sea anemone's tubular body. Any undigested food or waste is e_____ from the mouth.

Birds have a more complex digestive system with some unusual organs.

- The gizzard has sand inside to break up food.
- The crop is where birds store some food.
- The caeca is where eggs develop.
- The cloaca is the opening where eggs and waste pass through.

☞ **2** Use the word bank to label the diagram of a chicken's digestive system.



Word bank

intestine
crop
stomach
caeca
gizzard
cloaca

3 Label these statements true or false.

a) Birds have no teeth. _____

b) The gizzard has coarse sand inside. _____

c) Sea anemones don't have a digestive system. _____

d) The sea anemone's mouth is also its anus. _____

e) Sea anemones paralyse their prey before they eat it. _____

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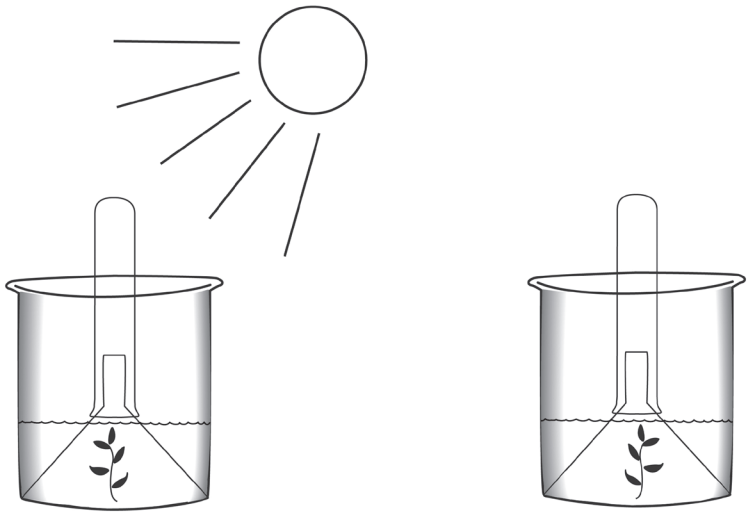
Photosynthesis

Two beakers of water each have an elodea plant inside them, with a test tube of water inverted onto a funnel sitting over the plant. One beaker is put in the sunshine. The other beaker is put in a dark cupboard. This is the **control**.



control

☞ **1** A day later, students need to test for oxygen. How do they do this? Number the steps below in the correct order.

___ Plunge the glowing toothpick into the open test tube.

___ Light a toothpick.

___ Remove the test tube from the water, keeping your thumb over the opening.

___ Blow out the flame.

___ Put the beaker on the bench.

2 Do the experiment yourself. What happened to the toothpick when you put it in the control test tube? _____

3 What happened when the toothpick was put into the sunlit test tube? _____

_____

4 How does this show that oxygen was present? _____

_____

5 Could the oxygen have come from somewhere else? _____

_____

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Teacher's notes – Transport systems

Achieve!

Objectives

- To recognise that groups of tissues form organs, each with specialised functions
- To understand that groups of organs work together as a system
- To know that gas exchange and nutrient delivery depend upon a transport system within a complex organism
- To recognise that different species may have different transport systems that are suited to their particular environment

Prior knowledge

Students should be aware of our lungs and heart and how they are vital to life. Students should also know that plants wilt without water and that water is carried up through the plant's roots.

Background

All cells need to exchange gases in order to survive. Complex organisms such as humans have a cardiovascular system that allows for gas exchange through the bloodstream. The heart maintains the movement of blood throughout the body and the lungs are the interface for gas exchange. Many marine organisms have different organs called gills that provide the same function. Plants have a simpler gas exchange process, however, water is needed in the leaves where photosynthesis takes place. The movement of water from the roots through the vascular bundles into the leaves is the basis of a plant's transport system.

Starter activity

Ask students to put these words into some context: oxygen, carbon dioxide, blood, lungs, heart, roots, veins, sap. Guide them towards describing typical transport systems in plants and animals.

Activity sheets

The Activity sheet 'The human heart' presents results from an experiment measuring pulse rates before and after exercise. Students graph the results and are then encouraged to do the experiment themselves and record and graph their own results.

The 'Gills vs lungs' Activity sheet demonstrates that gas exchange in aquatic organisms such as fish is different to that of land-based animals. Students are given two diagrams to label before answering some true/false questions.

The 'Putting down roots' Activity sheet illustrates how water and nutrients move through plants. Students are asked to order steps in an experiment. They are then encouraged to perform a similar experiment and to record their results by drawing their observations. Students will need to have grown a bean plant in sand prior to conducting the experiment. They will also need red ink or dye, a slide, a hand lens and a razor blade. Ensure close supervision when students are using the razor blades.

'Hydroponics' explains that many commercial crops, such as vegetables, are grown in this way. Students read a passage about hydroponics then answer questions to ensure they have understood the information. They finish off the lesson by completing a labelling exercise.

Extension ideas

Devise a bingo game using words to do with gas exchange and circulation in animals and water uptake in plants.

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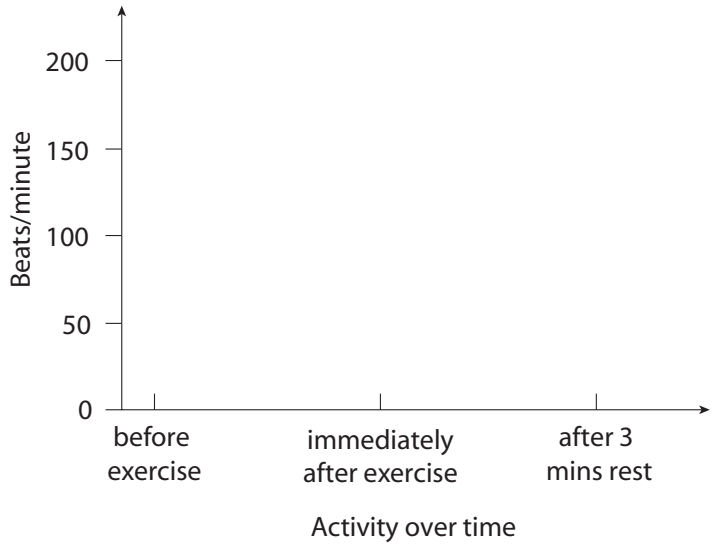
The human heart

Sophie took her pulse before she exercised. It was 70 beats per minute. She exercised for five minutes, then took her pulse again. This time it was 180 beats per minute. After resting for three minutes she took her pulse again. It was 90 beats per minute.

☞ **1** Plot Sophie's results on the graph below.



2 Working in a group, take your pulse rates before and after exercise. Record your results in the table below.

Name	Before exercise	Immediately after exercise	After 3 mins rest
	_____ beats/min	_____ beats/min	_____ beats/min
	_____ beats/min	_____ beats/min	_____ beats/min
	_____ beats/min	_____ beats/min	_____ beats/min

3 Graph your results in your workbook.

4 Why does your pulse rate increase during exercise? _____

_____

5 How are your results similar or different to Sophie's? Can you explain any differences?

_____

_____

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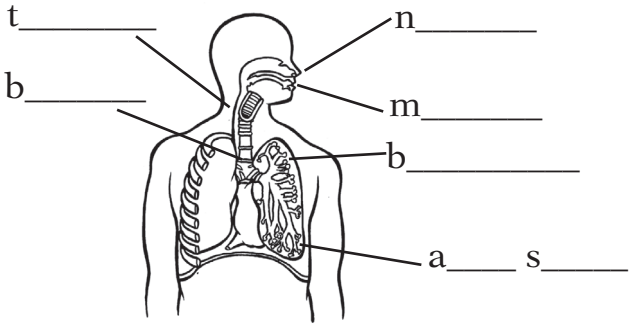
Gills vs lungs

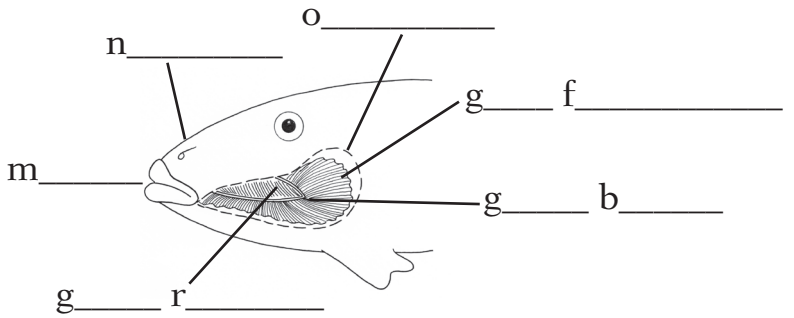
Fish and humans both need **oxygen** to breathe. Humans inhale air, which travels through the trachea into the lungs. Each lung is filled with tiny air sacs called alveoli, which transfer oxygen from the air into our **bloodstream**. This is exchanged for **carbon dioxide** which is **expired** when we breathe out. Fish take oxygen from the water when they open their mouths. Oxygen diffuses from the water into the gills and then enters the bloodstream.

☞ **1** Use the word lists to label the diagrams.



human lungs

trachea
bronchus
bronchiole
mouth
nostril
air sacs



fish gills

gill rakers
gill bar
gill filaments
mouth
nostril
operculum

2 Label these statements true or false.

a) Water pumps over the gills when a fish opens and shuts its jaws. _____

b) Humans do not have gills. _____

c) Expiration is when air enters through the mouth. _____

d) Human lungs have many air passages that end in alveoli. _____

e) Fish can suffocate on land. _____

f) Oxygen diffuses from the gill filaments into the water. _____

3 Rewrite the false sentences to make them correct.

_____

_____

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Putting down roots

☞ **1** Ravi wants to track the movement of water in plants. Below is a list of steps he needs to take to perform a simple experiment. The steps are jumbled up. Sort the steps into the correct order and rewrite them in the boxes below.

Draw the root specimen, showing which areas are red.

Place the seedling in a solution of red ink or dye.

Wash the sand away from the roots.

Cut the root into very thin slices with a razor blade.

Take a young bean plant that has been growing in sand.

Lay a slice on a slide and examine it with a hand lens.

Leave for a few hours until you see red dye in the leaves.

1

2

3

4

5

6

7

2 Now perform the experiment yourself, but take thin slices from the stem and leaf as well as the root. Draw what you see in each of the sections.

Root	Stem	Leaf

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Hydroponics

Read the passage below and then answer the questions.

Hydroponics is a way of growing plants without using soil. Some commercial vegetables, such as lettuce, are grown this way. Instead of garden beds, the farmer uses lengths of pipe filled with **nutrient-rich** water. The seeds are sown in moist coarse sand or synthetic pebbles and placed in little holes along the pipes. As the seeds germinate their roots go beneath the sand or pebbles and feed on the fertilised water. The water is changed regularly to keep the nutrient levels high. This method of growing plants keeps them free of weeds, and soil-borne pests and diseases.

☞ **1** In the paragraph above, circle the materials farmers use to grow plants using hydroponics.

2 Underline the words that describe what the plants feed on.

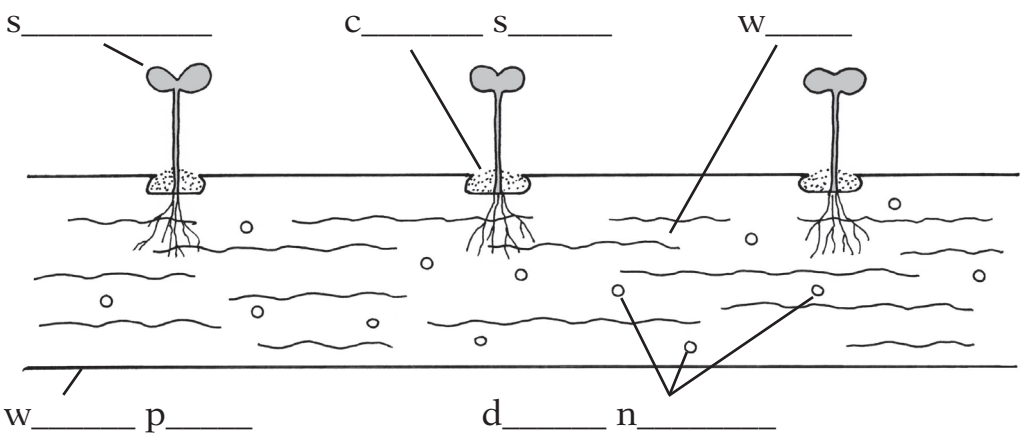
3 Why is the water changed often? _____

4 List two benefits of growing plants this way. _____

_____

5 Use the word bank to label the diagram below.

Word bank				
seedling	water	dissolved nutrients	water pipe	coarse sand



Growing plants using hydroponics

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Teacher's notes – Sensory systems

Achieve!

Objectives

- To acknowledge that different species have different sensory abilities to humans
- To identify parts of the human eye and ear
- To understand that plants can sense environmental conditions and react accordingly
- To recognise that our skin is our largest organ and has several roles

Prior knowledge

Students should be able to list the five senses. They should also be aware of medical and technological advances that can help when our senses are impaired, such as hearing aids, cochlear implants, skin grafts and eye surgery.

Background

The five human senses of touch, taste, sight, smell and hearing are common knowledge but animals and plants also have senses, many of which differ from our own. For instance, bats and dolphins use echolocation while other animals can recognise magnetic or electrical fields. Plants are sensitive to sunlight, temperature and day length.

Starter activity

Working in groups, blindfold one student per group and ask them to record their reactions to a range of stimulus, such as loud noises, strange smells and different textures. Discuss their responses. Talk about how taste can be diminished when our sense of smell is impaired. Ask students if they have ever been seasick and discuss contributing factors such as balance and body awareness.

Activity sheets

The Activity sheet 'The human eye' introduces the parts of the human eye. Students label a diagram and complete a crossword to become familiar with relevant terminology.

The 'Compound eyes' Activity sheet explains that insects have compound eyes. Students complete a cloze passage, followed by higher order questions to challenge students, and true/false questions to consolidate learning.

'Echolocation' introduces Australia's ghost bats – large carnivorous bats that use echolocation. Students answer questions and draw sound waves on a diagram to demonstrate understanding of how echolocation works. Higher order questions further direct students' learning.

In the 'The human ear' Activity sheet, students label a diagram of a human ear. The text introduces the hearing ranges of bats, dogs and humans, and follows up with some higher order questions.

The Activity sheet 'Following the sun' explains how plants react to stimulus such as a change in the amount of light. Students order the steps of an experiment and then predict the results.

'Skin and synthetic skin' presents a list of notes about synthetic skin. Students write a paragraph summarising this medical technology, then label a diagram to reinforce their understanding.

Extension ideas

Snails have no hearing and very poor eyesight, so they rely on taste, smell and touch as their main senses. Students could write a fictional paragraph, imagining they are a snail and talking about the things they encounter on their travels.

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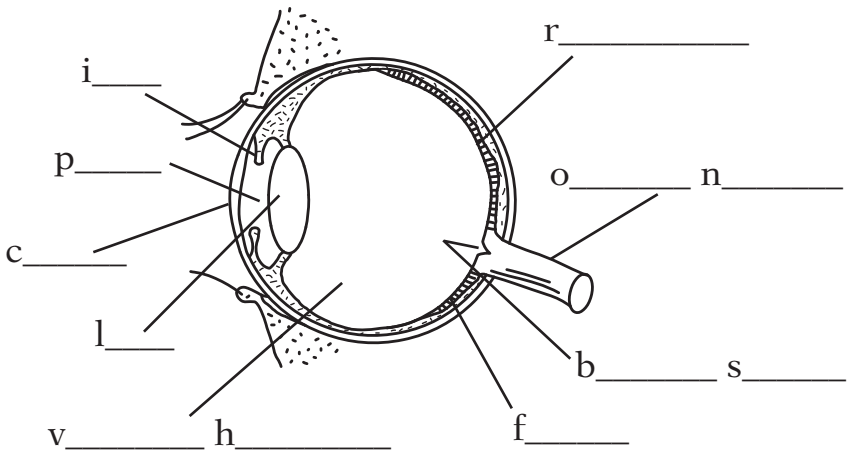
The human eye

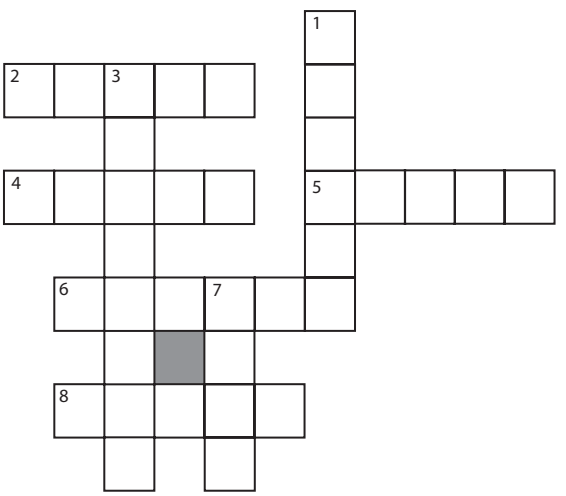
☞ **1** Label the diagram using words from the word bank.

Word bank

iris pupil lens retina optic nerve fovea vitreous humour cornea blind spot



2 Complete the crossword puzzle using the words below.



cornea
retina
nerve
iris
fovea
vitreous
optic
pupil

Across

2. Focal point on retina
4. _____ nerve
5. Optic _____
6. Layer of cells sensitive to light
8. Black spot in middle of eye

Down

1. Transparent disc in front of lens
3. _____ humour, liquid inside the eye
7. Disc with hole in the centre

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Compound eyes

☞ **1** Use the word bank to complete the passage below.

Word bank

lenses surface eyes above far sides insects
compound produce vision

Arthropods such as i_____ are the only animals that have **compound eyes**. Compound e_____ are made up of many small l_____. Each lens is positioned on the s_____ of a hemisphere. They can see a_____, below, behind, to the s_____ and straight in front, all at once. All these views p_____ a mosaic image, like the pixels in an enlarged photograph. The more lenses, the better the insect's v_____. The insect cannot see very f_____ – the c_____ eye is better suited to detecting movement, rather than detail.

2 Why do bees visit windblown flowers more than still flowers? _____

_____

3 Why is it so hard to catch a dragonfly? _____

_____

4 Why do grasshoppers have poorer vision than bees? _____

_____

5 Label these statements true or false.

a) Only arthropods have compound eyes. _____

b) Insects can see in many directions at once. _____

c) Compound eyes take photographs. _____

d) Compound eyes are good at detecting movement. _____

e) A compound eye has only one lens. _____

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Echolocation

Ghost bats live in northern Australia and are Australia's only carnivorous bat species. Ghost bats have long, large ears that are very sensitive to sound. They use **echolocation** to hunt at night for large insects, lizards, birds, frogs and small bats. Used by many bat species, echolocation involves using sound waves to locate food and find their way around in the dark.

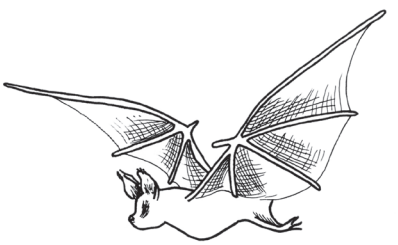
The sound waves bounce back from animals and objects around the bats. They listen for the returning waves to work out the size, location and direction of movement of animals nearby. If the waves return quickly, they know the animal is close by. If the sound waves are intense, this tells them that the animal is large. If the waves return first to the bat's left ear, then moments later to its right ear, this means the animal is on their left.

☞ **1** Describe a ghost bat's ear. _____

2 Why do ghost bats use echolocation? _____

_____

3 Draw sound waves coming from the ghost bat in blue and bouncing back from the bat's prey in red.



4 Where would the prey be if the sound waves return first to the bat's right ear?

_____

5 If the echoing sound wave is faint, what size would the animal be? _____

_____

6 If the sound waves take a while to return, how far away is the prey? _____

_____

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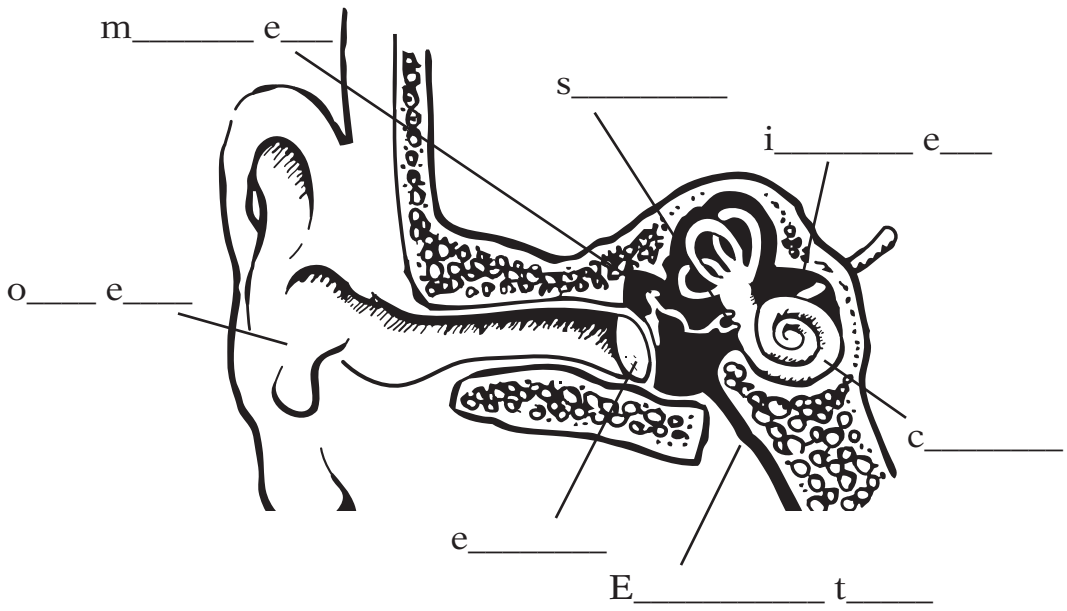
The human ear

☞ **1** Use the words in the word bank to label the diagram.



Word bank

outer ear
eardrum
middle ear
inner ear
cochlea
Eustachian tube
stapes

The table below shows approximate hearing ranges for humans, dogs and bats.

Species	Approx. hearing range (Hz)
Human	20–23 000
Dog	40–60 000
Bat	20–120 000

2 Why do bats have such a wide hearing range? _____

_____

3 Could humans hear the same sounds as bats? _____

_____

4 Explain what a dog whistle is, and why humans can't hear them. _____

_____

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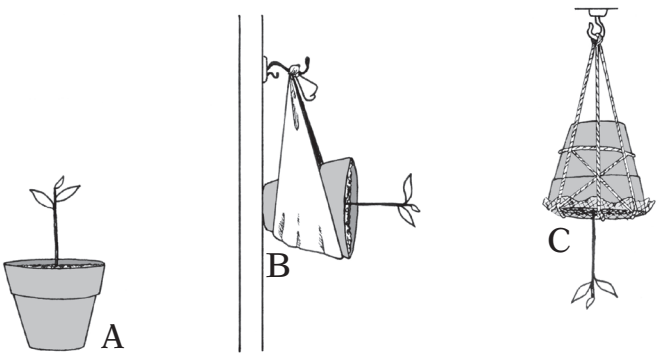
Following the sun

Plants are sensitive to light and actively seek it out. An experiment can be set up to prove that plants grow towards the light. Three pot plants can be placed in different positions, as shown below.



☞ **1** Draw how the plants would have grown after one week.

2 Rosa thinks that plants won't grow as tall if they grow in the dark, because they won't have enough light to grow quickly. How could she test this? Number the steps below into the correct order to design an experiment.

___ Record the results.

___ Put the rest of the seedlings in a dark cupboard.

___ Collect some seedlings.

___ Put two seedlings on the bench.

___ Water all the seedlings as needed.

___ Measure all the seedlings after one week.

3 Match the phrases below.

Shoots grow upwards	that is needed for photosynthesis.
Shoots grow towards light	water that is needed for plant growth.
Roots grow towards	to get more light and gases.

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Skin and synthetic skin

Synthetic skin

- Skin is the human body's largest organ.
- Severe burns damage large areas of the **dermis**.
- Large burns expose the body to infection and dehydration.
- The dermis doesn't regenerate but the **epidermis** does.
- Skin **grafts** are often rejected.
- Grafts don't provide enough replacement skin for large burns.
- Grafts cause scarring and don't stretch well.
- Artificial skin developed from animal collagen provides a new dermis.
- Epidermis grows over the artificial dermis.
- New skin grows as patient grows.

☞ **1** Read the notes above, then write a paragraph summarising the information about artificial skin.

_____

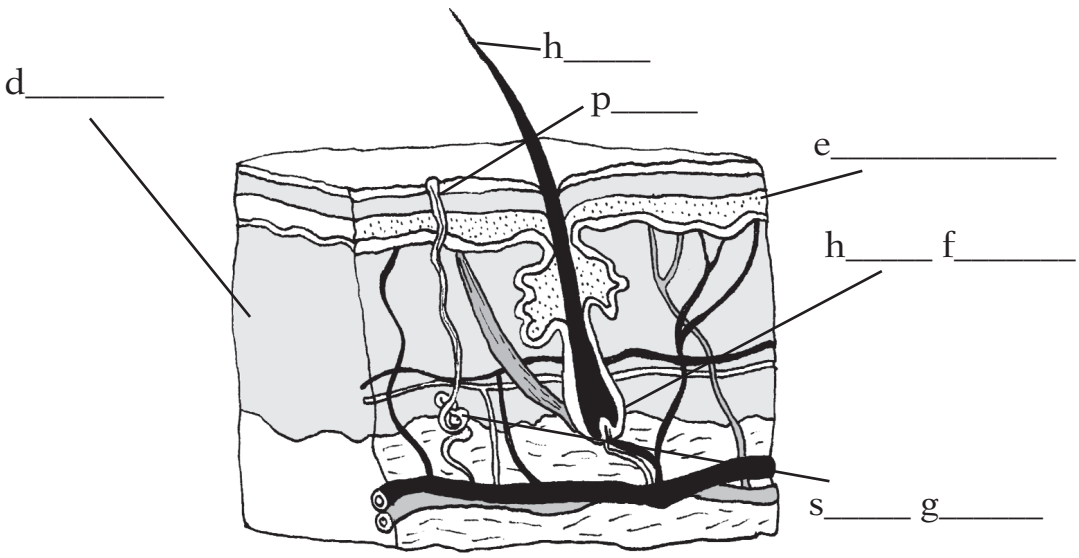
_____

_____

_____

_____

2 Use the word bank to label the diagram showing the layers of skin, below.



Word bank

dermis
epidermis
hair follicle
hair
pore
sweat gland

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Teacher's notes – Reproductive systems

Achieve!

Objectives

- To understand that in sexual reproduction, egg and sperm fuse to form a new individual
- To be able to recognise flower parts and their role in fertilisation
- To be aware of the difference between sexual and asexual reproduction
- To recognise that different species have different reproductive systems and behaviours

Prior knowledge

Students should know the structure of a cell and how it divides. Similarly, a fertilised egg begins to divide, grow and differentiate into a series of specialised organs which forms a new individual of a species. Students should be aware that reproduction of plant species can be via a new material, such as a seed, or through other methods, such as bulb division or cuttings.

Background

Students should know the basics of mammal reproduction, where genetic material is passed from the male to the female via sexual intercourse. In marine animals such as fish and sea horses the reproductive process may be different. Reproductive behaviour in some species, such as the kangaroo, can be dependent on environmental factors. Many plants have the ability to reproduce both sexually and asexually. ***Teachers should refer to their own school's sex education policy prior to delivering this unit.***

Starter activity

Introduce the topic of reproduction with the perennial question: what came first, the chicken or the egg? Why do species such as frogs and fish, which lay hundreds of eggs, not care for their young? Discuss how the concept of 'egg' can also be applied to mammals, and humans in particular.

Activity sheets

In the Activity sheet 'Human reproduction', students complete a crossword and match statements as a revision exercise for any previous work done on human reproduction.

The Activity sheet 'Red kangaroo reproduction' provides information about the reproductive life of the red kangaroo and explains that adult female kangaroos are always pregnant. Students complete a cloze passage, then fill in a timeline detailing the birth and nurturing of joeys.

'Asexual reproduction' introduces the idea that some organisms are able to reproduce on their own. Starfish can reproduce in a similar way to the way plants reproduce from cuttings. Students summarise notes about this process to demonstrate their understanding. They then match a set of phrases to consolidate their learning.

The Activity sheet 'Flowering plants' introduces phrases about flowering plants for students to match. Students then complete the labels on a diagram.

The 'Gender roles' Activity sheet explains that it's not always the female that gives birth and nurtures the offspring. This Activity sheet looks at alternatives to female reproductive roles. Students complete a cloze passage, true/false statements and a timeline exploring the way barramundi change gender as they age and the role male seahorses play in birthing their young.

'The malleefowl' introduces the unusual way this solitary Australian bird takes an active role in incubating its eggs. Students examine the reproductive habits of this mound builder, and complete a labelling exercise to consolidate their understanding.

Extension ideas

Have students investigate the gestation periods of a range of animal species, such as elephant, cow, blue whale, mouse and dog. How do these differ to hatching times for turtle eggs, chickens, etc? This information can be tabulated to display in the class.

Selected answers

Red kangaroo reproduction

Jan: Joey W is born;
Mar: Joey X is born;
May: Joey X is ejected.

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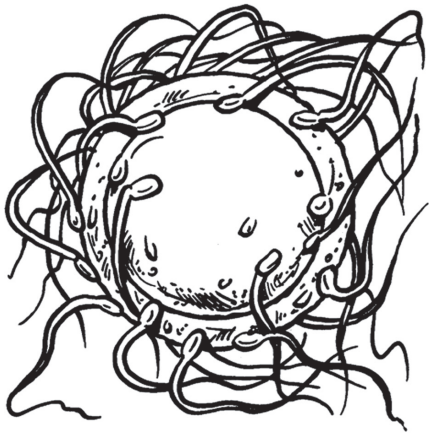
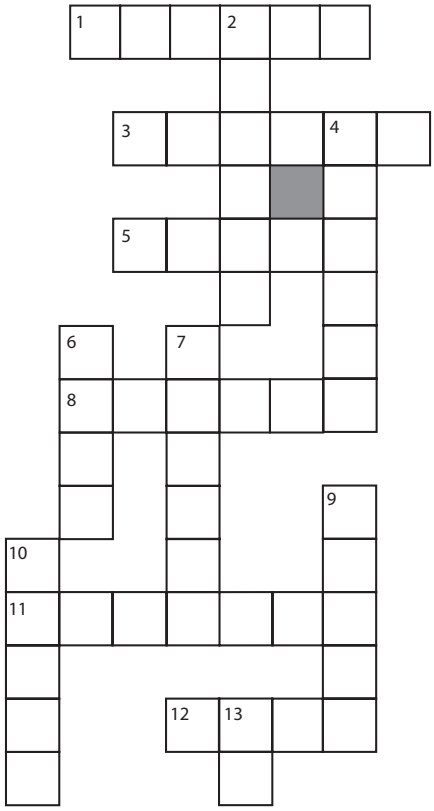
Human reproduction

☞ **1** Complete the crossword using the words in the word bank.





Word bank

testes
sperm
month
uterus
cervix
duct
penis
VF
ovulate
vagina
zygote
sexual
ovum

Across

1. Opening of the uterus
3. Fertilised egg
5. Male sexual organ
8. Where the foetus develops
11. Discharge egg from ovary
12. Egg

Down

2. Birth canal
4. Sperm are produced here
6. The ovum travels down the ovi_____
7. Copulation: _____ intercourse
9. Semen contains these
10. An egg is released each _____
13. In-vitro fertilisation: I_ _

2 Match the statements below.

The uterus grows	opening of the uterus.
The cervix is the	approximately once a month.
Sperm is made	as the foetus develops.
Menstruation occurs	in the man's testes.

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Red kangaroo reproduction

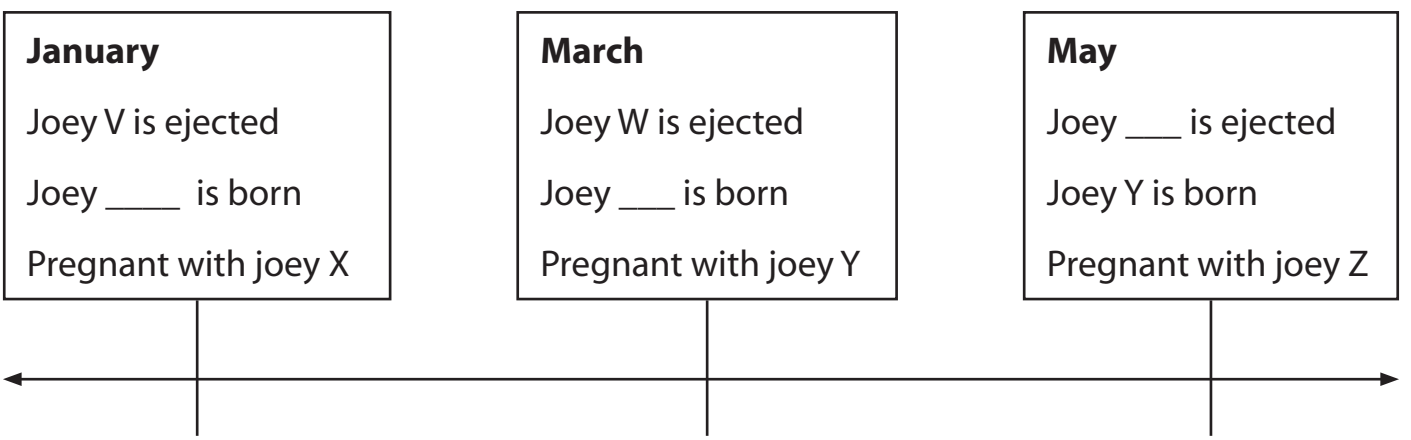
Word bank

nipples
first
reproductive
stage
weighs
joey
pouch
drink
different
ejected
pregnant
development
droughts
feeding

☞ **1** Use the word bank to complete the passage below.

The female red kangaroo has three stages to her r_____ cycle. The f_____ stage is mating. The pregnancy lasts 33 days, but d_____ may stop for months without any harm to the foetus. The second s_____ is birth. When the red kangaroo is **born** it is the size of a raisin and w_____ just under 1 gram. It crawls into the mother's p_____ and attaches itself to one of her n_____. The third stage is ejecting the j_____ from the pouch for good – although it may still tuck its head into her pouch for an occasional d_____. During d_____ a joey may be **ejected** before it reaches maturity. As soon as the joey has been e_____ the female mates straight away to make sure she is p_____ again. This means kangaroos are always **pregnant** and f_____ one joey, or even two at d_____ stages of development.

2 Look at the timeline below showing three cycles of pregnancy for a female red kangaroo. Fill in the blanks to show the stages for joeys V, W, X, Y and Z.



Months in the life of a female red kangaroo

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Asexual reproduction

- The cut stem grows roots.
- Doesn't grow from seed.
- Many plants can reproduce **asexually**.
- Keep the same characteristics as the mother plant.
- **Cuttings** are lengths of stem placed in soil.
- Doesn't need bees or ants to pollinate.
- Starfish can also grow from a large piece of severed arm.

☞ **1** Use the notes above to write a paragraph about asexual reproduction.

_____

_____

_____

_____

_____

2 Match the statements.

A new starfish can grow	doesn't need a partner.
Many flowering plants	stem that starts to grow roots.
Asexual reproduction	identical to the mother plant.
The daughter plant is	can grow from cuttings.
A cutting is a length of	from a large severed arm.

3 Add the missing vowels (a, e, i, o, u) to complete the sentences below.

To take a c__tt__ng, first s__l__ct a strong sh__ __t.

Cut the shoot d__ __g__n__lly just __b__v__ a bud.

D__p the c__t end in honey or powdered r__ __t hormone.

Tr__m the l__ __v__s so there __r__ very few.

Put th_ cutting into pr__p__r__d soil and w__t__r well.

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Flowering plants

Flowers are the reproductive structures of many plants.

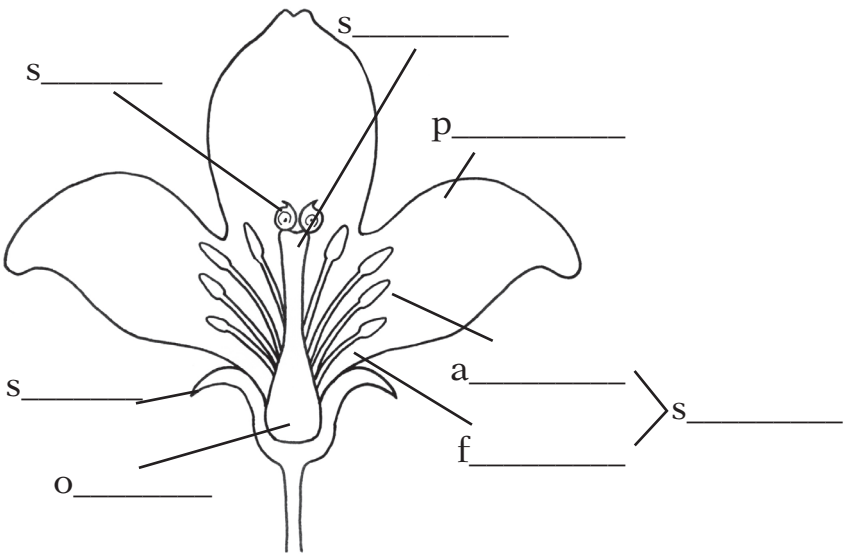
☞ **1** Match the phrases below about flowers.

Many flowers contain male	pollinators of flowers.
Bees and ants are common	the male part of a flower.
Some species have male flowers	petals and sepals fused.
Pollination is when pollen is	from the **fertilised** flower.
Fruit and seed form	transferred from flower to flower.
The **stamen** is	parts and female parts.
Some flowers have their	and female flowers.

2 Use the words in the word bank to label the diagram of a flower.

Word bank

petal	style	stamen	anther
filament	stigma	ovule	sepal



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Gender roles

☞ **1** Use the word bank to complete the passage below.

Word bank					
maturity	fish	move	females	freshwater	Australia

Barramundi are f_____ that live in rivers, creeks and mangrove estuaries in northern A_____. They start life as males in the upper reaches of f_____ rivers and streams. They reach m_____ at around three to four years old, then m_____ downstream to coastal waters for spawning. At around five years old they change gender and become f_____.

2 What gender would the barramundi be at each point along the timeline below?

Months in the life of a barramundi

3 Use the word bank to complete the passage.

Word bank								
mate	males	become	bigger	birth	fertilises	unusual	female	birthing

Seahorses are unusual in the animal kingdom because it is the males that b_____ pregnant. They also choose to m_____ with only one partner, which is u_____ in fish species. The females are generally b_____ and more colourful than the m____ in order to attract a mate. Males have a b_____ pouch located below their chest area. The f_____ deposits her eggs inside the male's pouch, where he f_____ them. After two to three weeks he gives b_____ to the young.

4 Circle 'true' or 'false' for each of the following statements.

a) Male seahorses are bigger than female seahorses. True/False

b) The female deposits her eggs in the male's pouch. True/False

c) The female gives birth after three weeks. True/False

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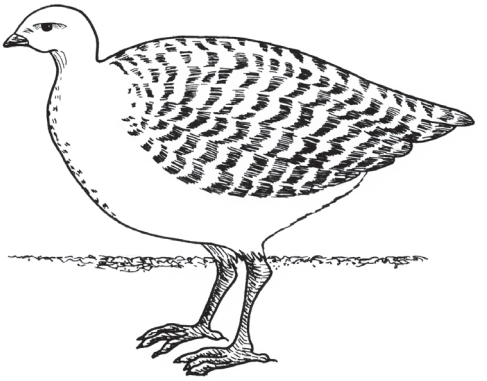
The malleefowl



The malleefowl is an Australian ground-dwelling bird. They are shy, solitary birds who only come together to mate. The male malleefowl digs a big hole in the ground and fills it with organic material such as sticks, bark and leaves. He then waits for rain. Once the organic matter is damp, he covers it with a thick layer of sand. The organic matter begins to decompose and generates heat. Every day the male checks the mound's temperature by poking his bill into it. When it is warm enough, the female lays her eggs inside the mound. Then she leaves the site and the male bird tends the mound daily, maintaining the internal temperature at 33°C. He does this by scraping away or adding the insulating sand.

The eggs take 7–9 weeks to incubate. By this time the male bird has stopped maintaining the mound and has left the hatchlings to fend for themselves. The chicks hatch inside the mound and struggle their way out of the sand. They never know their parents.

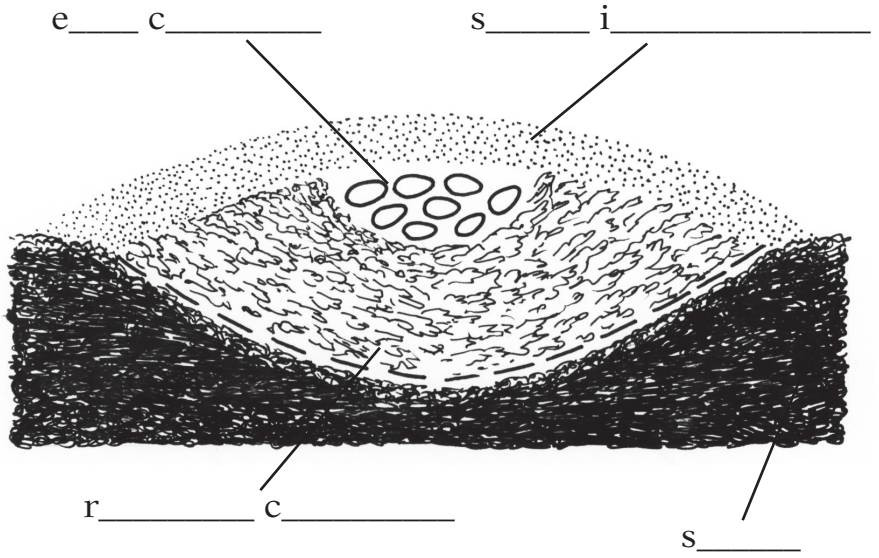
☞ **1** Read the text above, then underline the part that describes how the nest warms up.

2 Circle the temperature the nest needs to be kept at while the eggs incubate.

3 List the materials the male malleefowl uses to make his nest. _____

_____

4 Use the word bank to label the diagram.



Word bank

egg chamber
soil
rotting compost
sand insulation

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Topics include

- Classification basics
- Classification
- Food chains
- Food webs
- Looking at cells
- Cell structures
- Energy systems
- Transport systems
- Sensory systems
- Reproductive systems

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